Radio Frequency (RF) Energy Harvesting Ultra-Low-Power Transceiver for Environmental Sensing

Midterm Document
EEL4914 - Senior Design I
Group III

University of Central Florida
Department of Electrical and Computer Engineering

<u>Group Members and Majors</u>
Ryan Witt, EE
Jacob Henderson, EE
Nicholas Vakhordjian, EE/CPE

<u>Project Sponsor</u>
Dr. Reza Abdolvand
UCF Dynamic Microsystems Lab

<u>Project Reviewers</u>
Dr. Hakhamanesh Mansoorzare | Dr. Justin Phelps
Dr. Xun Gong | Dr. Zakhia Abichar

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2.0 Project Description

2.1 Motivation and Background

Environmental monitoring systems are an essential part of industries, cities, and research institutions delivering real-time data about temperature, humidity, air quality, and other environmental aspects. Traditional sensors rely heavily on batteries or active components, which limits the flexibility of deployment, increases maintenance costs, and dramatically impacts their long-term use. In remote or hard to access environments, replacing or recharging batteries is impractical, making conventional power solutions more challenging for large scale or long-term sensing applications. This obstacle motivates the exploration of alternative power sources that can enable autonomous, maintenance-free operation.

Radio Frequency (RF) energy harvesting has emerged as a promising solution to this limitation. Ambient and intentionally transmitted RF signals can be captured and converted into a usable DC power supply, enabling electronic systems to operate without direct connection to a power source. Recent advances in rectifier efficiency, methods of rectification, impedance matching networks, and low power microcontrollers have made it feasible to design sensor nodes that function solely on harvested RF energy. By utilizing RF energy harvesting, sensing devices can achieve indefinite operational lifetimes, reduced environmental impact, and greater implementation capabilities. This approach aligns with the growing demand for sustainable, self-powered electronics in the Internet of Things (IoT) ecosystem.

This project aims to design and implement an ultra-low-power environmental sensing transceiver (sensor node) capable of operating entirely from harvested RF energy. The system integrates an RF energy harvester, a high efficiency power management stage, an ultra-low-power microcontroller, and a 2.4 GHz Bluetooth low energy wireless transceiver optimized for intermittent, energy aware communication. By demonstrating reliable temperature and humidity sensing within a 1–3 meter RF power transfer range, this project showcases the feasibility of battery-less wireless sensor nodes. The resulting platform provides a foundation for future applications in sustainable IoT networks, distributed environmental monitoring, and autonomous embedded systems.

2.2 Prior Related Work

2.2.1 Industrial Thermocouples

A common solution to industrial environmental temperature sensing is thermocouples. These electrical devices contain two materials with dissimilar thermal properties, which creates a temperature-dependent voltage [1]. As such, the voltage across the thermocouple increases with temperature. Commercial thermocouples are widely available and are inexpensive, and unlike other temperature sensing methods, they do not require an active power source to create a difference in potential.

2.2.2 RFID Temperature Sensor Tags

Passive RFID temperature sensors are a similar product available on the market already. They utilize a temperature-dependent resistor (TDR) to sense the ambient temperature using power gathered during the scan [2]. These sensors are small, cheap, and completely passive, but their range is limited to centimeters due to the high order drop-off of power transfer in the near field region through resonant-inductive coupling. This project aims to increase the scanning distance and scope of sensing available compared to these types of products by operating in the far field region, where power drops off at a much slower rate.

2.2.3 Wireless Ultra-Low-Power Sensor Platform for Environmental Monitoring

A group at the Hanover University of Applied Sciences designed an RFID-based environmental monitor sensor with a range of around 3 meters. The system was intended to operate on a duty cycle of 10% and take around 1 minute to charge and send data back to the transceiver. The sensor was designed to operate on an input power of at minimum -15dBm, whereas a typical RFID tag requires at least –6dBm to operate. This paper offers a good starting point for sensor selection. As the fundamentals of RFID are vastly different from those of Bluetooth-Low-Energy, the MCU choice does not directly map to this application [3].

2.2.4 Advances in Passive Wake-up Radios with Wireless Energy Harvesting for the Internet of Things applications

A study published by the National Library of Medicine describes the passive wake-up of radios via energy harvesting. The article describes the importance of duty cycling and considers it crucial for the lifetime of a passive electronic device [4]. It reduces energy consumption due to listening in an idle state. The radio state will be in sleep mode and is switched to active mode when its internal synchronization clock is ready to transmit or receive data. During RF energy harvesting, electromagnetic waves are converted into electricity using a rectifying antenna, or rectenna. Energy propagation occurs through electromagnetic radiation which covers frequencies ranging from 3 kHz to 300 GHz [4]. Optimal power delivery and reduced RF losses occur only when the antenna is impedance matched to the rectifier diode's input stage.

2.3 Goals and Objectives

As the world of RF technologies continues to expand, many new opportunities for unique and innovative products with the possibility of transforming the way we communicate, interact with the world, and interact with each other, emerge. This project exists to serve as an example of this growth and the ways RF technologies can be applied to the world, evolving into the powerful, scalable technologies that drive modern communication, sensing, and intelligent systems.

Shown in table 1, we begin by specifying our team's goals for this project. These goals exist to define a scope for our team's focus for this project. At the same time, aside from the implementation of our goals, the top priority of this project is to serve as an educational and academic template for future aspirants in the field of electrical and computer engineering.

Table 1.1: Project Goals Overview

Goal	Description
Basic	• Create a compact, handheld RF transceiver to wirelessly power a passive sensor station. • Develop a passive sensor station PCB with a temperature and humidity sensor that relies only on incoming RF energy to power onboard electronics. Sensor data is then transmitted back to the handheld transceiver to be shown to the user.
Advanced	• Increase effective transmission range between the handheld transceiver and passive sensor station. • Increase scope of sensing to include more robust, efficient, and/or complex sensors. • Utilize ambient radio waves to passively charge the sensor without the use of the main board antenna.
Stretched	• Implement automated polling of the sensor to allow for remote data acquisition from a distance. • Expand the scope of the project to focus on IoT integration with multiple passive sensor boards. • Utilize a more efficient means of RF energy rectification.

Table 1.2: Project Objectives Overview

Objective	Description
Basic	• Develop energy harvesting circuitry using a RF to DC converter, a capacitor, and voltage regulating ICs to supply power to the microcontroller and sensors. • Develop a high-power transceiver board to send RF power to the passive sensor using a highly directional antenna to minimize power usage and receive Bluetooth transmissions for display.
Advanced	• Improve regulator efficiency to decrease the amount of time RF exposure is required before the sensor node can activate

	• Design more effective (more efficient, higher gain, more directional) transmission and reception antennas which are not possible to buy from a supplier for reasonable prices. For example, a patch antenna array with higher directional gain. • Add additional sensors to the board without significantly increasing the required power consumption.
Stretch	• Lower total cycle energy usage to the point where ambient 915MHz ISM can generate sufficient energy. Other ambient RF signals, such as TV and radio, can be designed around for energy harvesting as well. • Write a software stack that can determine the location of a given sensor and direct the antenna to send energy towards it. This will aid in integrating multiple passive sensors in an interconnected array of IoT devices. • Implement an acoustoelectric MEMS device as a replacement for diode rectification in the RF-DC converter.

The primary objective at the foundational level is to demonstrate a passive RF-powered sensing node capable of operating solely from beam-harvested energy. This begins with establishing a reliable energy-harvesting antenna built with circuitry that can convert RF energy into a stable DC supply. The system must accumulate this energy in components such as a capacitor or a battery and reliably start under realistic RF power levels. Once power is available, the goal is to operate an ultra-low-power microcontroller that wakes briefly, samples the humidity and temperature using a sensor, transmits the measurement, and returns to deep sleep. The sensor must be chosen and integrated with strict attention to its energy cost. Another RF transmission will send the data to a receiver, where the measurements will be displayed. At this stage, the main objective is to manage a simple energy-aware cycle: harvest, wake, measure, transmit, and sleep, while avoiding energy over-use.

At the advanced level, the project shifts towards efficiency and reliability. The receiver should operate with a harvesting efficiency of at least 60%, meaning that 60% of the power received by the antenna should be turned into usable energy. Maximizing this value directly increases the effective range of the system and lowers the time required to ping each station. The antennas of the transmitter and receiver should operate within 1 MHz of each other to increase transmission efficiency and thus increase effective range. Additional sensors may be added to increase the functionality of the receiver circuit. These, however, will draw additional power. These could thus decrease effective range and must be counterbalanced by the methods described previously.

The stretch objectives begin to include aspects that could be defined as research-grade. The main addition would be the inclusion of an acoustoelectric device that would replace the diode rectifier. This would remove the large inherent loss of diode forward voltage drop and could significantly increase the effective range. Acoustoelectric devices are inherently complex, costly, fragile, and under large amounts of active research, which makes their inclusion a stretch goal. A second large goal is the development of an array of multiple RF-powered sensors, each capable of sharing the same RF signal channel through strategies such as time slotting. Power transmission occurs in the 915MHz ISM band. As such, there may be a large amount of available power in the ambient environment for our receiver to capture. A second stretch goal would be to energize the receiver using this ambient power, periodically poll the surroundings, store this in memory, and transmit all stored data when pinged. A large amount of software optimization may go into this aspect of the stretch goals, which could be a stretch goal itself. An example of possible implementation would be an algorithm that detects available RF power and limits polling frequency respectively.

2.4 Features/Functionalities

The proposed system of our project incorporates a high-efficiency RF energy harvesting front end designed to capture and convert low-power RF signals into stable DC energy suitable for autonomous operation. This subsystem includes a voltage doubling rectifier optimized for gigahertz frequencies, a carefully tuned impedance-matching network to maximize power transfer, and an energy storage capacitor bank that will store the energy required for the circuit to function. Together, these components enable the device to operate even under weak RF field strengths, extending its usability across a wide range of deployment environments by vastly expanding operational range. The power management unit further enhances functionality by providing cold-start capability, over-voltage protection, and intelligent energy routing to ensure that sensing and communication tasks only occur when sufficient energy is available.

At the heart of the device is an ultra-low-power microcontroller engineered for intermittent, energy-aware operation. The MCU interfaces with environmental sensors, such as temperature, humidity, or air quality modules and performs data acquisition when pinged with minimal energy overhead. Its firmware is designed around a deep sleep dominant duty cycle, waking only when the energy storage capacitor reaches a predefined threshold. This adaptive approach allows the system to dynamically adjust its operational range based on real-time energy availability, ensuring reliable performance even when RF power levels fluctuate. Additional features such as onboard ADCs, low-power timers, and configurable wake-up interrupts further enhance the MCU’s ability to operate efficiently within the constraints of harvested energy.

Complementing the energy harvested sensor module is a handheld transceiver unit, which provides both the RF power signal and the data display. The handheld device emits a controlled continuous wave RF signal in the 915 MHz ISM band to energize the sensor module within a 1-meter range. In addition to power delivery, the handheld unit includes a receiver that captures the sensor’s transmitted data packet once the sensor module has harvested enough energy to wake, measure, and communicate. A microcontroller within the handheld transceiver manages transmission timing, and data display, making the handheld device the central controller for field testing, validation, and real-time environmental data retrieval.

The stretch goal of this project will feature a novel acoustoelectric RF-DC rectifier. This rectifier will increase the overall efficiency of the power harvesting section by removing the forward voltage drop inherent to all diodes. This acoustoelectric addition also improves over a design using MOSFET rectifying circuits, as they are active and thus require power before they can rectify. This results in a chicken/egg problem for the application that a fully passive acoustoelectric device will not have.

2.5 Engineering Specifications

The system and component requirements for the energy harvesting sensor and handheld transceiver are based upon the marketing requirements listed in Table 3. These marketing requirements are criteria which the project deliverable should meet to satisfy the stated problem and be usable as a product. Some requirements, such as communication between the sensor and transceiver, are system-level, whereas other requirements are specific to either the sensor or handheld device. Table 4 lists the system and component level specifications which will meet the proposed marketing requirements. The highlighted items are those which will be used to demonstrate project functionality during the demo.

Table 1.3 - System and Component Specifications

Marketing Requirement	Engineering Requirement	Justification
1, 2, 4, 5, 6	1. The usable charging distance should be up to 1 meter.	The wireless sensor will be separated from the handheld transmitter. The sensor must be wirelessly powered across this measurable distance and transmit data back to the main board.
4, 5, 6	2. Transmitter EIRP must not exceed 36 dBm.	Per FCC Title 47, Section 15.247, the maximum allowable directional antenna gain is 6dBi when using a 1 Watt (30dBm) of power.
6	3. Transmitter must utilize unrestricted frequency bands.	The FCC regulates which frequency bands can be used. Commonly used bands for low power and short-range devices include 915 MHz and 2.4 GHz.

1, 5, 6	4. Sensor charge time should be ≤ 30 seconds.	The handheld transceiver must first wirelessly charge the sensor before any sensor readings can be taken and sent back. Our team believes this is a reasonable amount of time to wireless charge over distance given FCC restrictions on transmitter power level.
1, 2	5. Sensor board electronics should not exceed 100 μJ.	The sensor will be passive, meaning no battery is present. As such, the sensor electronics must consume minimal energy.
4, 5	6. Transceiver battery should last at least 3 hours.	The handheld transceiver will utilize a screen to display sensor data to the user. The screen is expected to stay on while RF power is sent intermittently to the passive sensor. Since the amplifier will not always be on, the handhelds battery should last a long time.
3	7. There should be ≥ 2 sensor readings visible on the transmitter.	The transmitter board should display any sensor readings that the receiver board generates. The receiver board should measure at least two different aspects of the environment.
Marketing Requirements		
1. The sensor should be battery-less. 2. The sensor electronics should consume a low amount of power. 3. The sensor should be capable of acquiring multiple types of data. 4. The handheld transceiver should be capable of transmitting a high-power RF signal to charge the sensor. 5. The handheld device should have a long battery life. 6. The system devices should be FCC compliant.		

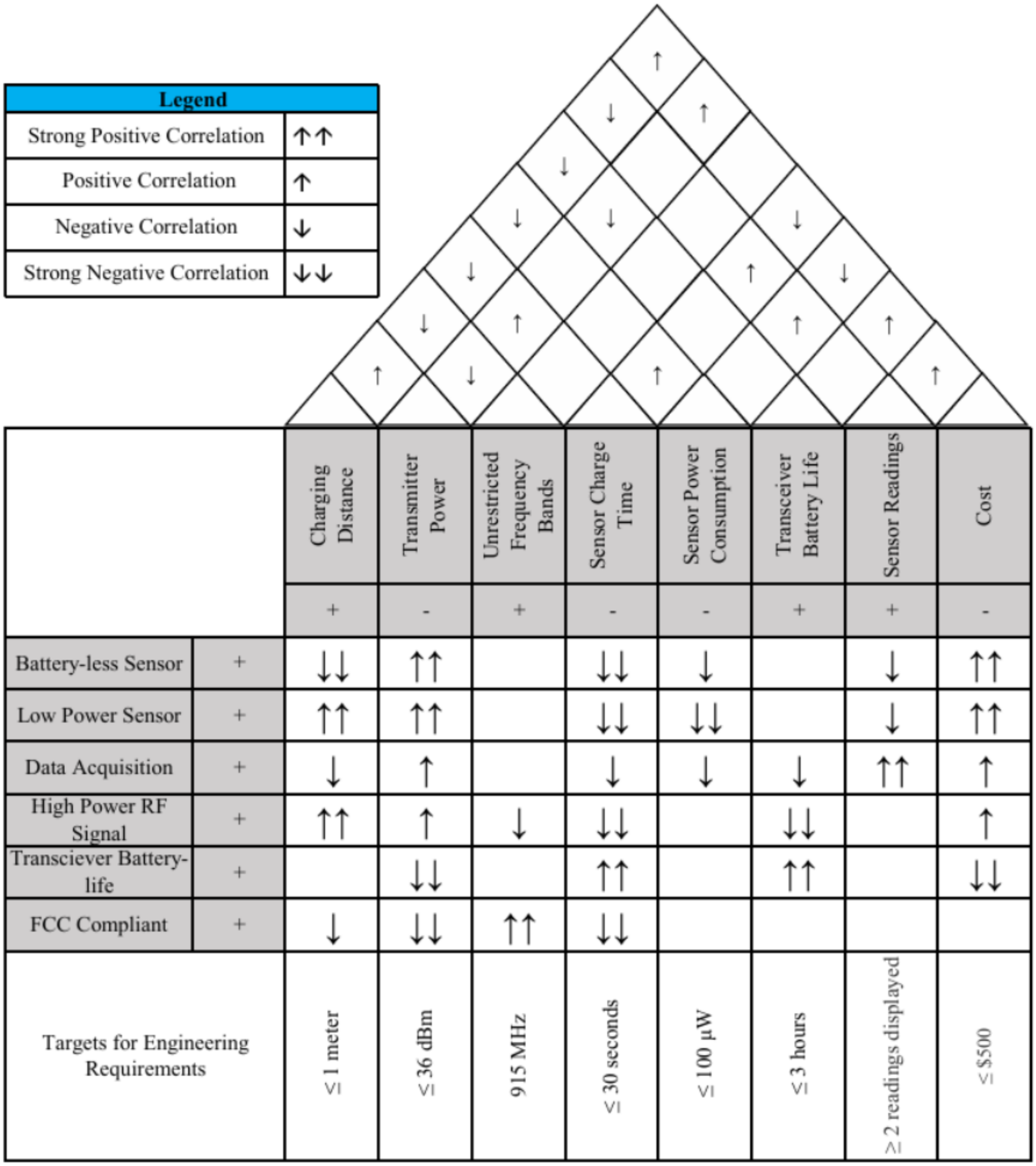

Figure 1.1 - House of Quality by Authors

2.5 Hardware Block Diagrams

The hardware diagram provides a high-level view of the major components that make up the RF energy harvested environmental sensing system and the handheld power transceiver. It illustrates how the handheld transceiver, energy harvesting circuit, microcontrollers, and sensing elements are connected to enable wireless power delivery and data acquisition. By outlining these relationships, the diagram helps clarify how energy flows through the system and how each module contributes to overall device functionality.

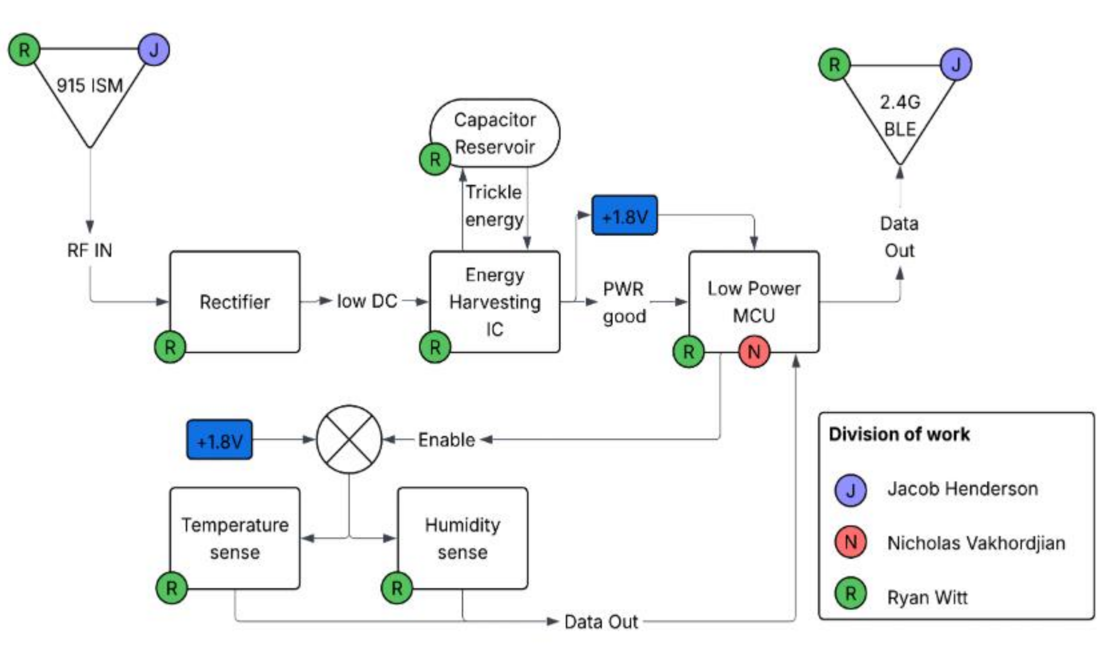

Figure 2.1 – Energy harvesting sensor block diagram by Authors

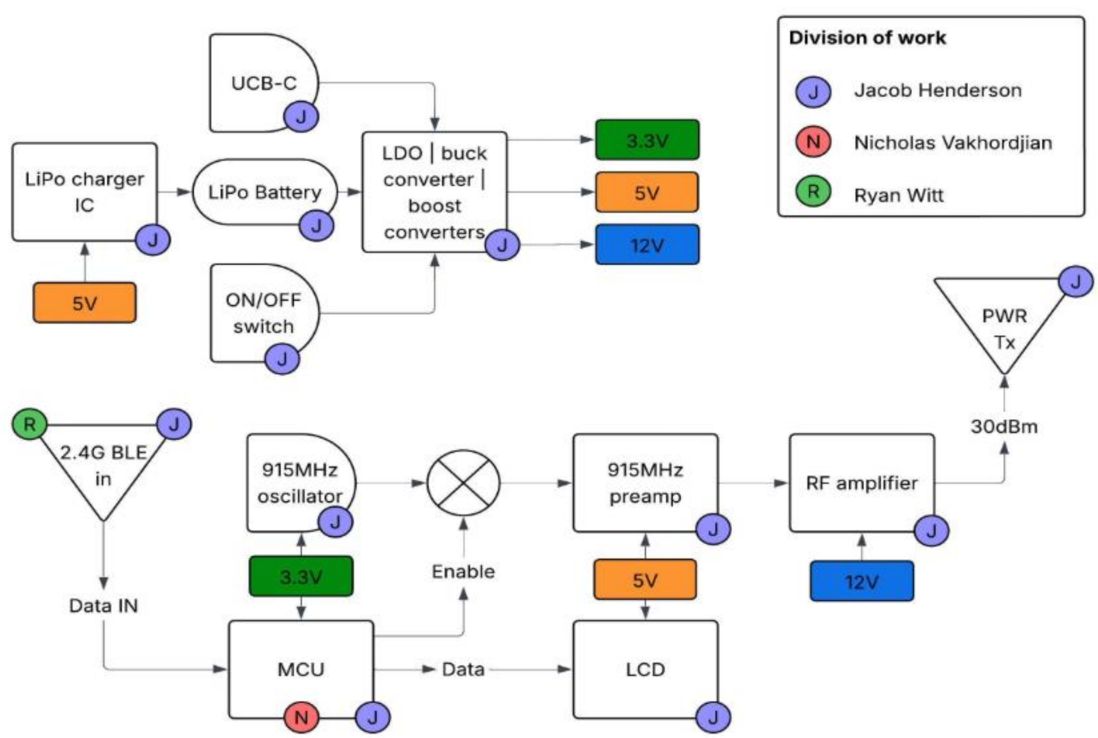

Figure 2.2 – Handheld transceiver block diagram by Authors

2.6 Software Flow Charts

The diagrams below represent a high-level software flow governing how the RF energy harvested environmental sensing system operates. Once the handheld transceiver is powered on, it begins by transmitting an amplified RF power signal toward the sensor board. This RF signal provides energy to the harvesting circuit on the sensor node, allowing the capacitor to accumulate enough charge to activate the microcontroller and sensing peripherals. As soon as the handheld transceiver begins transmitting RF power, it simultaneously enters a listening state, waiting for incoming sensor data packets. On the sensor board, once sufficient energy is stored, the device wakes from sleep mode, performs a quick initialization, samples the environmental parameters, and transmits a compact RF data packet back to the handheld device. If the handheld transceiver successfully receives this packet, it decodes the information and displays the measured temperature and humidity to the user. If no data is detected, the handheld transceiver triggers an error, informing the user that no data has been received. In other words, indicating either insufficient harvested energy, a communication issue, or a sensor fault. This process allows both devices to coordinate smoothly, enabling the sensor to operate solely on harvested RF power while still delivering environmental data.

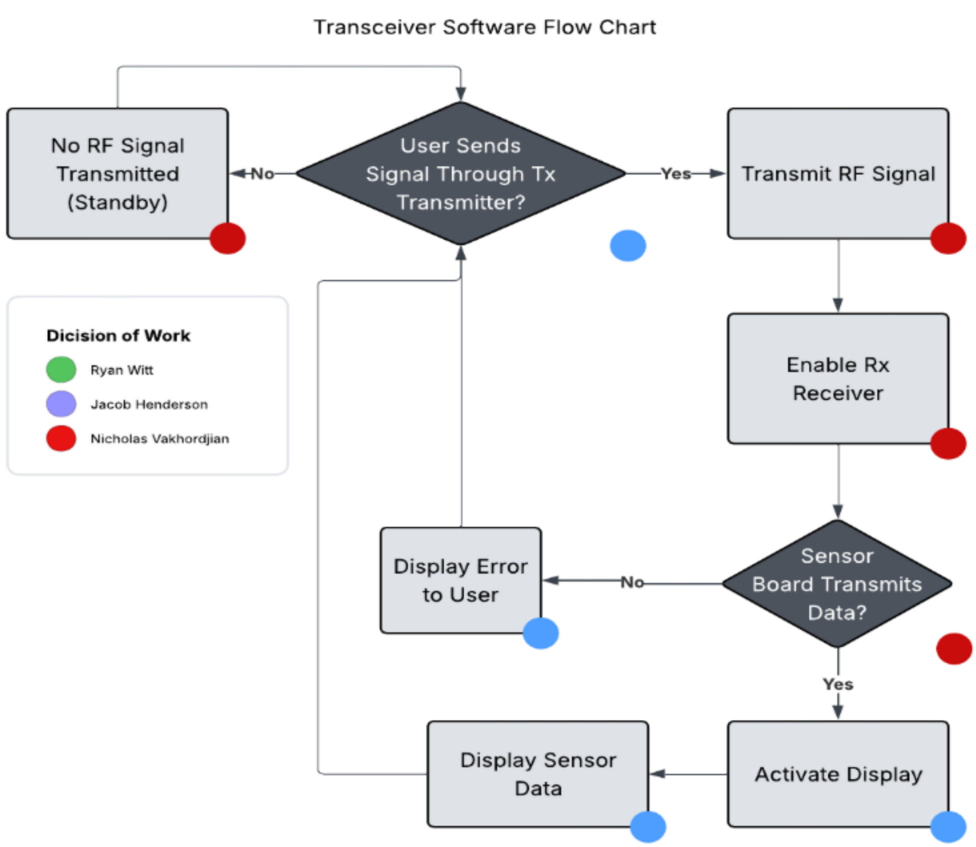

Figure 3.1 – Transceiver software flow chart by Authors

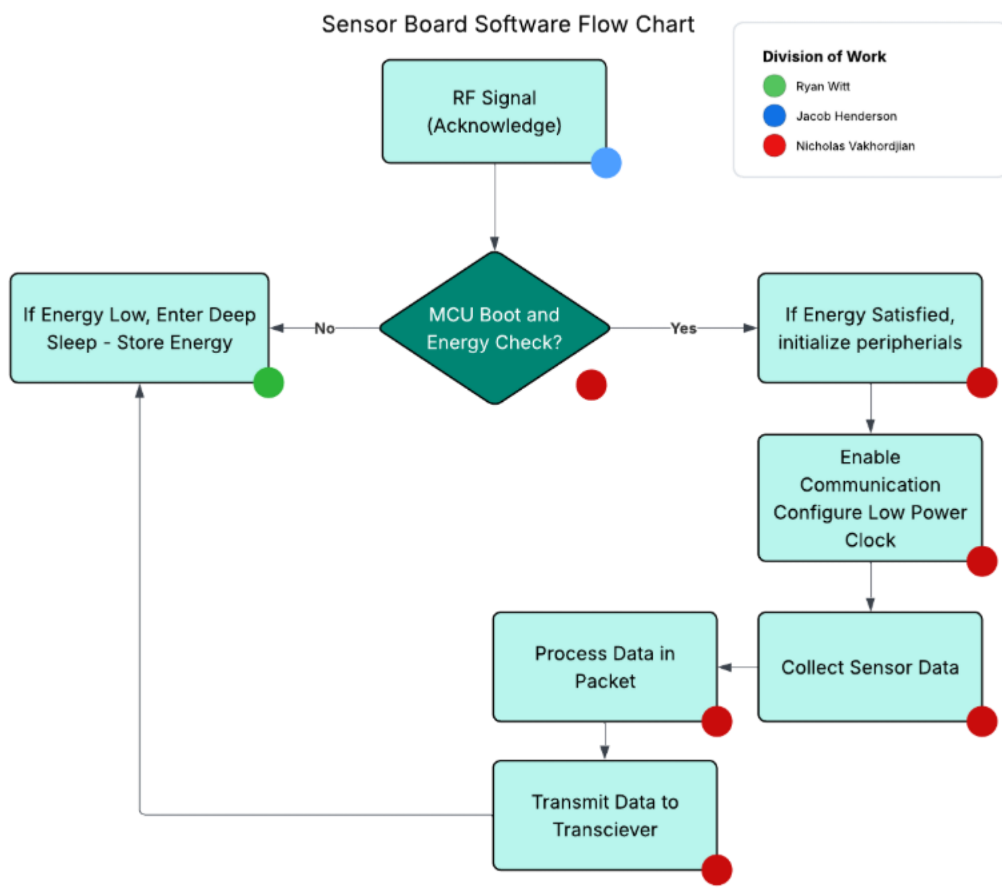

Figure 3.2 – Sensor software flow chart by Authors

2.7 Project Prototype Illustration

An illustration of our prototype is included with this Divide and Conquer document to convey our initial design ideas. The handheld transceiver and energy harvesting sensor will each require their own PCB; thus, two separate prototype device boards will be developed. Figure 1 shows a system level diagram of the handheld transceiver, including its high-power antenna (left), Bluetooth receiving antenna (right), user display, ON/OFF switch, transmit button, power electronics, RF electronics, and microcontroller. Information regarding the interconnection of blocks can be found in 2.2 Handheld Transceiver.

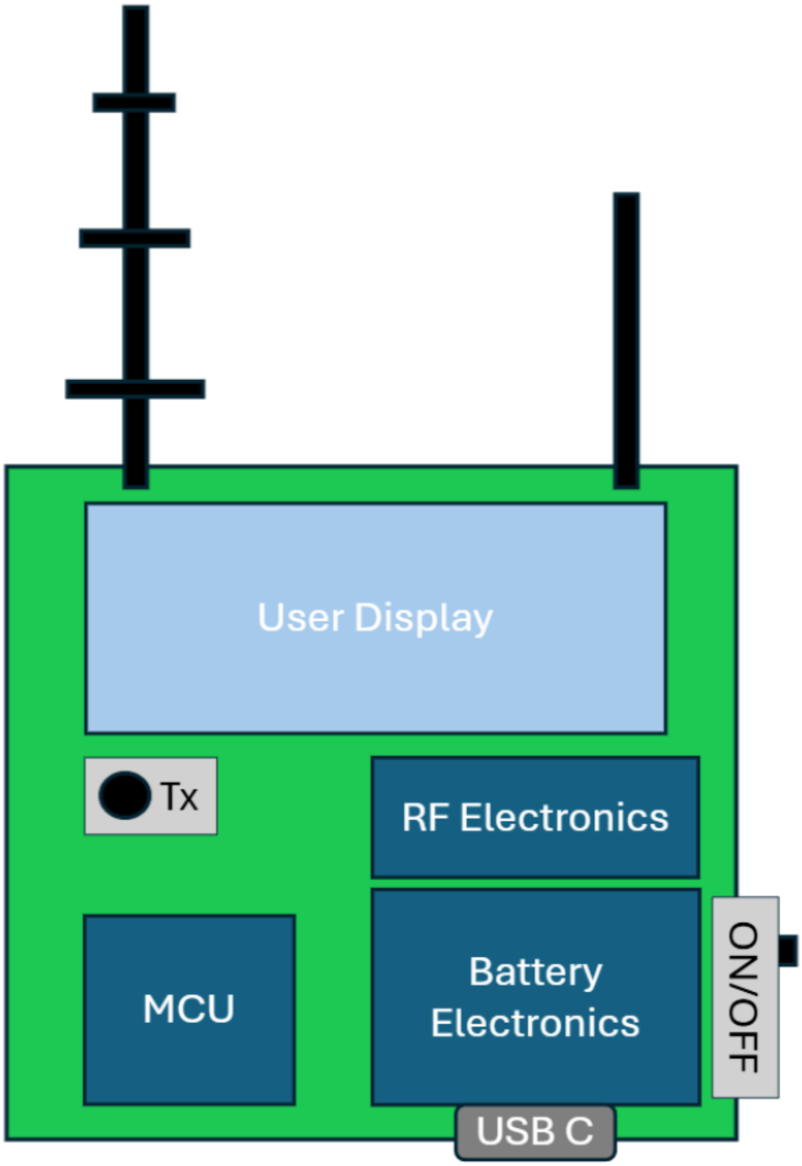

Figure 2.1 - Handheld transceiver prototype illustration by Authors

The energy harvesting sensor will be a separate board and will require its own prototyping. Two antennas will be needed, one to harvest RF energy and the other to send data back to the handheld transceiver. Harvested RF energy must be rectified into DC before the onboard electronics can utilize it. The large supercapacitor in the middle of the prototype is kept as close to the low power MCU to minimize loss of power across long traces. Information regarding the interconnection of each block can be found in 2.1 Passive Energy Harvesting Sensor Node and Figure 2.2 shows this initial prototype PCB which we plan to design and build.

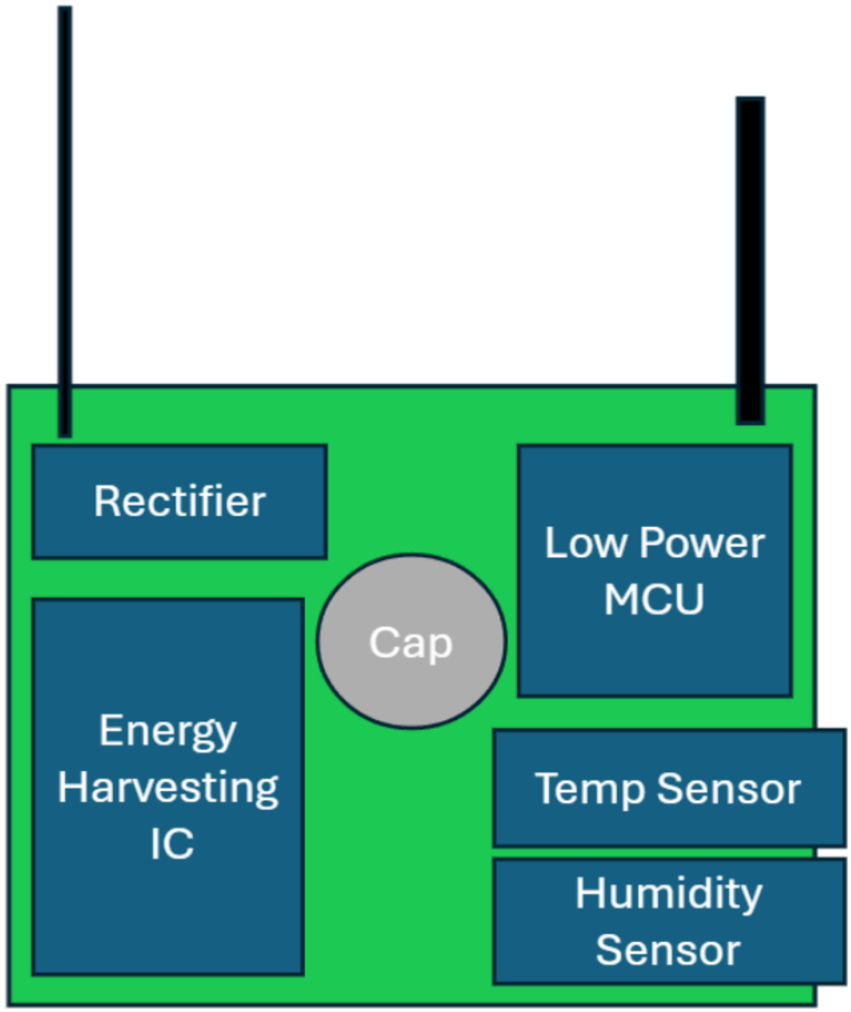

Figure 2.2 - Energy Harvesting Sensor Prototype Illustration

3.0 Research and Comparison of Technologies

(45 – 60 pages) *Should be in order of hardware, communication protocols, then software.*

3.1 Processing Hardware

The handheld transceiver and energy harvesting sensor both require embedded processing to function; however, each PCB faces different constraints. The handheld contains a battery and multiple peripherals which allow the user to intuitively interact with the device, while the sensor faces strict low-power requirements. Both devices must be capable of using wireless communication to transmit and receive data.

3.1.1 Energy Harvesting Sensor

The most important aspect of the energy harvesting sensor is the total energy usage of each ping. As such, the most important aspect of all parts constituting the energy harvesting sensor is their total “one-shot” energy usage. This will be the major deciding factor when differentiating between available technologies and their corresponding products. After energy usage is cost. A lower cost product with similar one-shot energy usage will be selected over one with a higher cost. Ease of implementation is third. Ease of implementation is defined in this project as ease of soldering, number of additional components required, and ease of programming.

This system will be power-gated; Therefore, “sleep” or “shutdown” quiescent currents are generally of little importance (except for in the case of the microcontroller). Start-up time, start-up current, or more accurately the combination of these parameters in the form of total start-up energy, is far more important in the context of this application.

3.1.2 Handheld Transceiver

The handheld transceiver will be used to interact with the energy harvesting sensor. It is responsible for transmitting RF energy to charge the sensor and receiving sensor data wirelessly from the passive sensor board. Unlike the energy harvesting sensor, the electronics on the handheld transceiver are not limited to the same ultra-low power requirements due to the presence of battery power.

The data received wirelessly from the sensor should be recorded and displayed to the user on the handheld device. Hardware for displaying the data and input peripherals is necessary so the user can read the data and decide what should be done with it. For example, the data may be saved to a cloud server and displayed on a website.

3.1.3 Microcontrollers

Microcontrollers (μCs or MCUs) are small, single chip computers that are readily available to consumers for use in embedded systems. These chips contain processors, memory, data input/output, and A/D hardware that interfaces with peripherals. The processor in the microcontroller is responsible for executing any code which has been programmed to memory by the user. The inclusion of memory within the microcontroller is convenient since it does not have to be added externally. This proves useful for embedded applications or prototyping projects due to everything being contained within an enclosed package. However, the size of memory is limited due to it being contained within the microcontroller’s package. In applications where more memory is needed, such as high computation computing, microcontrollers will not have enough memory to be a viable option.

The low cost of microcontrollers makes them ideal for many applications, such as consumer electronics, industrial applications, and Internet of Things (IoT) devices. In addition, microcontrollers have many general purpose input/output (GPIO) pins which the programmer can use when interfacing with external hardware. Microcontrollers are also easily programmable since they use common high-level languages (C, C++, Arduino) and require little hardware-level programming due to containing all computing components on the chip. Overall, the all-in-one package, many peripheral options, and ease of programming make microcontrollers a great choice for embedded applications where data from multiple external sources must be processed.

3.1.4 Microprocessors

Microprocessors (MPU) are central processing units (CPUs) on a chip. Unlike microcontrollers, these chip processors do not have peripheral interfaces built into the front end of the package. As such, they cannot directly interface with common peripherals without using additional external hardware to interface with the MPU. While microprocessors have cache memory within them, they do not have random access memory (RAM) or electrically erasable programmable memory (EEPROM) built within the package. This means external memory must be added and interfaced to store the executable instructions when designing around them.

What MPUs lack in peripheral capability, they greatly make up for in pure computing power. Their chip space is instead used for complex architectures that include many computing cores, each capable of executing their own instructions. This makes modern microprocessors multi-threaded and catered towards general-purpose applications instead of low-power or embedded devices.

3.1.5 Field-Programmable Gate Arrays (FPGA)

FPGAs are integrated circuits which contain reconfigurable logic blocks that can each operate in parallel. The majority of pins on an FPGA are GPIO, making them both highly customizable and capable of interacting with many peripherals. The reconfigurable logic blocks within an FPGA can be programmed to act as front ends interfaces – such as ADCs, DACs, serial communication interfaces, and voltage drivers – which allows for very complex hardware implementation, and since each logic block can operate in parallel, each peripheral can be communicated with at the same time which gives FPGAs far more throughput and parallel processing power than MCUs and MPUs.

FPGAs see applications in signal processing and real-time systems due to their customization. However, their high parallel computing ability makes them consume large amounts of power. Furthermore, coding FPGAs requires the use of low-level description languages such as Verilog and complex verification techniques to verify the timing and logical outputs of configured blocks.

3.1.6 Comparison of Embedded Computing

When comparing the available options for computational hardware in our project, the number of peripherals that can be interfaced and overall power consumption must be considered. Complexity of implementation is also an important factor, as it increases the amount of hardware and software required to make work. Among the options, microcontrollers provide the best solutions to the listed criteria in Table 3.1. These chips have built-in peripheral and memory components and are easily coded using high-level languages. This makes their complexity and cost of implementation relatively low when compared to the MPUs and FPGAs. Most importantly, microcontrollers consume the least amount of power compared to the other options, which makes them ideal for low-power energy harvesting and battery powered devices. While the computing ability and parallelism of MCUs are lower than its counterparts, it is a necessary tradeoff to ensure power consumption of the device is kept to a minimum.

Table 3.1 - Comparison of Processing Hardware

Criteria	MCU	MPU	FPGA
Power Consumption	Low	Low	High
Computing Ability	Moderate	Moderate	High
Parallelism	Limited to None	Moderate	High
Coding Complexity	Low	Moderate	High
Hardware Integration	Inherent peripheral support	No internal memory or peripherals	Must be configured
Hardware Cost	Very Low	Low to Moderate	Moderate to High

3.1.7 Microcontrollers for Energy Harvesting Sensor

As described earlier, the single most important aspect of microcontroller selection is the total energy used per ping. Due to the power-gated nature of this application, we may divide the operation of a µC into TWO sections. The first section is system start-up. This section is defined as the period where low-level hardware is being initialized before application code is executed. At a high level, this section includes the initialization of system clocks, crystals, memory, and cores; the stabilization of power systems, reference rails, oscillators, and pin voltages; and the loading of application code from system rom into cache and memory. The second section is code execution. This section is defined as the period where application code is being executed. Code execution can be divided into FOUR sections. The first section is I2C command / ADC read. During this period, the microcontroller sends commands to sensors on the I2C bus and reads analog values from sensors whose output is an analog voltage and stores them in memory. The second section is sleep. During this period, the µC enters an ultra-low-power mode while it waits for sensors on the I2C bus to read values and store them in memory. The third section is wake-up. This period begins when the I2C slaves transmit their return message. The microcontroller will wake up from its ultra-low-power sleep mode and store the received values in system memory. The fourth section is transmit. During this period, the microcontroller arranges the values stored in memory, prepares them for a Bluetooth low-energy (BLE) transmission, and transmits the values read over BLE.

The energy usage of all microcontrollers will be analyzed for each of these sections, and the microcontroller with the lowest total energy usage will be selected for use in the application. Due to the engineers' familiarity with STM32 µCs, they will be selected if possible; However, multiple brands of µCs will be investigated for usage.

Table 3.2 – List of µCs and their power characteristics

Brand	**µC**	**Calc. Efficiency (CM/mA) (µA/MHz)**	**Sleep quiescent (uA)**	**Transmit efficiency 0dBm (mA)**	**Comment**
Nordic	NRF54LV10A	123 CM (1.5v) 246 (3veq) 32 CM (1.5v) 16 (3veq)	1.7 (1.5v) 0.85 (3veq)	10.1 (1.5v) 5.1 (3veq)	1.2-1.7V operation
	NRF54LS05B	250 CM (3v) 15.625	0.7 (3v)	4.8 (3v)	
	NRF54L05	193 CM (3v) 21	0.9 (3v)	4.8 (3v)	
	NRF54LM20A	193 CM (3v) 18.75	1.0 (3v)	5.0 (3v)	
Si. Labs	EFR32BG22	27 (3v)	1-1.7 (3v)	3.4 (3v)	
Renesas	DA1435	29 (3v)	1-2.7 (3v)	3.9 (3v)	Calc eff: CPU running code from ram running on XTAL32 OSC at 16 MHz, DC-DC off
	DA1433	29 (3v)	1-1.7 (3v)	3.9 (3v)	↑ ↑ ↑
TI	CC2340R	53 (3v)	0.7	5.3 (3v)	
	CC2640R2F	48.5 CM 61	1	6.1	
STM	STM32WB05	14 (3.3v) 15.4 (3veq)	0.6-1.15	4.3 (3.3v) 4.7 (3veq)	CHOSEN MCU
	STM32WB10	90 (3.3v) 100 (3veq)	0.3(?)-3	8.6	Sleep quiescent is

					hard to determine
	STM32WB55	90 (3.3v) 100 (3veq)	0.3(?)-3	8.6	↑ ↑

The Nordic Semiconductor line of low power Bluetooth transceiver ICs represent an incredibly competitive selection of microcontrollers for this task. Their most impressive aspect is computational efficiency. The NRF54LS05X microcontrollers operate a 128MHz ARM Cortex-M33 processor at less than 16 µA/MHz at 3V. With a microscopic 700nA sleep quiescent current, this microcontroller has the ability to immediately wake up, spend minimal time initializing itself, sending I2C commands, and complete analog reads, and enter a nearly powerless sleep state. Total energy consumed by the receiver board is defined by $E = V * I * t$. By minimizing t, total energy consumed by the receiver can be vastly reduced. The only microcontroller that approaches this efficiency is the STM32WB05, but it runs at half the frequency. While the STM may be more efficient while operating alone, it is not the only IC in this circuit. All other ICs in this circuit have their own off-state quiescent currents, and they are therefore also consuming power while the MCU is operating. A faster MCU therefore may in fact create a more energy efficient circuit overall despite being slightly less efficient in its computation. That being said, it is incredibly difficult to determine which MCU would create a circuit with lower total energy usage. Typical sensor quiescent consumptions are on the order of microamps to nanoamps, whereas a running MCU can consume on the order of milliamps. At this scale, the higher efficiency of the STM chip could vastly overcome the energy loss of a longer total duration. Where the NRF54LS05X falls short is in its Bluetooth transmission efficiency. While good, it does not reach the level of microcontrollers designed by Silicon Labs or Renesas. Therefore, depending on the size of Bluetooth transmission, the Silicon Labs or Renesas chips may in fact outperform the NRF54LS05X. For the small amount of total data that is being transmitted in our case, total Bluetooth transmission time will be on the order of microseconds and therefore constitutes only a very small portion of the overall operating time. As described before, small time is equivalent to small energy. Overall, the NRF54LS05X is one of the best chips available for this task. It is, however, slightly outclassed by the STM32, as will be discussed. Therefore, this microcontroller has not been selected for use in this project.

The Silicon Labs EFR32BG22 offers an interesting advantage. It has the single lowest Bluetooth transmission power of researched microcontroller chips at 3.4 mA. This chip has a transmission power so low that it in fact surpasses the reception efficiency of many microcontrollers that were researched. It has a middle of the pack sleep current and has neither bad nor good calculation efficiency. For applications where large amounts of Bluetooth data must be transmitted, and therefore large amounts of time are dedicated to BLE transmission, this microcontroller could outcompete the others in energy usage based on its transmission efficiency alone. However, this project will only be transmitting a small amount of Bluetooth data, and therefore this microcontroller is not well-suited for this project. Additionally, the authors are not familiar with Silicon Labs microcontrollers and are further unfamiliar with their suite of microcontroller programming products. Therefore, we have elected to not use this microcontroller for this project. The time required to learn the Silicon Labs suite of products was deemed to be futile.

The Renesas DA1435 and DA1433 are similar to the Silicon Labs EFR32BG22 in most regards, only slightly worse. They have a slightly worse computational efficiency of 29 μA/MHz vs the Silicon Labs computational efficiency of 27 μA/MHz, their sleep quiescent current is the same at 1μA, and their Bluetooth transmission efficiency is undoubtedly good at 3.9 μA but is still severely outclassed by the Silicon Labs 3.4 μA 0-dB transmission efficiency. In nearly every aspect, the Renesas microcontrollers are either equivalent to or outclassed by the Silicon Labs EFR32BG22. Furthermore, the EFR32BG22 is not well suited for this project. If the Renesas microcontrollers are essentially an inferior version of the Silicon Labs microcontrollers which in the first place are a poor choice for this project, then there is no reason to choose them. It is for these reasons that the Renesas DA1435 and DA1433 have not been selected for use in this project.

The Texas Instruments CC2340R and CC2640R2F were researched because of the legendary status of the Texas Instruments MSP430 suite of ultra-low power microcontrollers. It was assumed by the authors that TI, a bastion of high-quality electronics products, would have a competitive low-power Bluetooth transceiver microcontroller. At the disappointment of the authors, it was discovered that the Texas Instruments suite of low power Bluetooth microcontrollers are less than ideal for this project. Save the high-power STM32WB35/55 microcontrollers, the Texas Instruments “low power” microcontrollers have the worst computational efficiency of researched microcontrollers at 58μA/MHz and 61 μA/MHz for the 2340R and the 2640R2F respectively, almost 4 times worse than the highest efficiency STM32WB05. Their sleep quiescent currents are good (2340R) and average (2640R2F) at 0.7μA and 1μA respectively, which should be expected from TI. Sleep current was the selling point of the original MSP430. Their transmit efficiency is again, save the high-power STM32WB55/35, the worst of researched microcontrollers at 5.3 mA and 6.1 mA for the 2340R and the 2640R2F, respectively.

The ST Microelectronics STM32 series of ultra-low-power microcontrollers is likely the single most popular line of microcontrollers available on the market today. They generally have exceptional sleep-mode energy usage, are incredibly computationally efficient, have a wide breadth of support documentation (official and unofficial), and have an impressive suite of development software. The STM32WB series of microcontrollers is the ST Microelectronics dedicated line of Bluetooth-low-energy capable microcontroller integrated circuits. Included in the researched list are two of the authors previously used microcontrollers, the STM32WB55/35. These are higher-power MCUs not optimized for ultra-low-power operation and thus have the worst computational efficiency, probable sleep current, and transmit efficiency of researched microcontrollers. However, the ST Microelectronics dedicated ultra-low-power Bluetooth-low-energy capable STM32WB05 is an entirely different story. The computational efficiency of the STM32WB05 is the best on the list at 15.4 µA/MHz, narrowly edging out the Nordic Semiconductor NRF54LS05B and its 15.625 µA/MHz. Similarly, its 0.6 µA quiescent sleep current is the best of researched microcontrollers. Again, it narrowly beats the NRF54LS05B and the Texas Instruments CC2340R, both of which have quiescent sleep currents of 0.7 µA. The only aspect of this chip that is not the best of researched microcontrollers is its Bluetooth transmission efficiency. It has a transmission efficiency of around 4.7 mA at 0 dBm and is therefore bested by the Renesas and Silicon Labs microcontrollers. However, given the fact that this application will not be transmitting large amounts of Bluetooth data and will as such be spending a relatively small amount of time in this higher current transmission mode, the higher transmission current is an acceptable compromise. It is for these reasons that we choose the STM32WB05 microcontroller for our application. It outright beats all other researched microcontrollers in the two particularly important spaces of computational efficiency and sleep quiescent while still having an acceptable Bluetooth-low-energy transmission efficiency.

3.1.8 Microcontrollers for Handheld Transceiver

Everything stated regarding the choice of microcontroller for the sensor holds true for the transceiver device as well. The microcontroller should be energy efficient and have high computational ability. Since the microcontroller will perform most of its tasks during charging, transmitting, and receiving – all of which occur within quick succession – the microcontroller should also have a low sleep quiescent current. With these factors in mind, the most straightforward choice is to choose a microcontroller from the STM32 family of devices, as these have been shown to have exceptional computational performance, strong Bluetooth compatibility, and very low sleep currents. However, unlike the sensor board, the transceiver's microcontroller must be capable of driving peripherals and handling interrupts while simultaneously transmitting power and then receiving data. As such, the chosen STM32 microcontroller should have two cores: one dedicated to primary tasks and interacting with peripherals and another to handle wireless Bluetooth communication.

Of the options in the STM32WB family, the STM32WB55 proves to be the best all-around choice. This microcontroller hosts two cores, with one being dedicated to wireless protocols. This ensures the wireless Bluetooth communication between the transceiver board and sensor board can occur continuously without fear of transceiver board peripherals needing to temporarily pause their actions. The main core is a One Cortex-M4 which contains an FPU with Adaptive Real-Time Accelerator that can run up to 64 MHz. The other core is a One Cortex-M0+ which has less functionality and is dedicated to running radio and Bluetooth communication tasks. The STM32WB55 comes in a 48-pin QFPN package with a total of 30 GPIO pins, and natively supports SPI, UART, and I2C communication protocols. The microcontroller also supports modern USB connectivity, which provides a useful alternative for software debugging and power delivery if the need arises.

3.5 Temperature and Humidity Sensors

Temperature and humidity sensors are critical subjects in our environmental sensing project. Choosing the sensor with the best characteristics of system performance is vital in determining its efficiency in an energy harvesting application. Monitoring thermal and moisture levels, these sensors enable adaptive power management strategies and provide essential environmental data for applications such as smart electronics, industrial automation, and IoT projects. Integrating low power and high accuracy sensing devices is therefore fundamental to maintaining stability within the energy harvesting architecture.

3.5.2 LMT70 Analog Temperature Sensor

The LMT70 is a small analog temperature sensor designed by Texas Instruments for applications requiring accurate, low power temperature measuring. The device uses a CMOS analog architecture that produces a linear voltage output proportional to temperature, with a typical slope of -5.19 mV/°C. This linearity simplifies signal conditioning and makes the sensor easy to interface with standard ADCs in microcontrollers. With a supply voltage range of 2.0 V to 5.5 V and a maximum supply current of only 12 µA, the LMT70 is ideal for energy constrained systems.

One of the defining features of the LMT70 is its accuracy across harsh ranges of temperatures. The sensor achieves ±0.05°C for typical accuracy between 20°C and 42°C, making it suitable for medical equipment, human body temperature, and some regions for environmental monitoring. Beyond this range, the device maintains strong performance, with maximum errors of ±0.2°C from -20°C to 90°C and ±0.36°C from -55°C to 150°C. This broad operating range allows the LMT70 to serve in industrial, environmental, and IoT applications where both low and high temperatures require reliable measurements.

The LMT70 also includes an output enable pin. This pin allows multiple sensors to share a single ADC channel by enabling or disabling their outputs as needed. In multi-sensor systems such as distributed IoT nodes, this capability reduces system complexity, simplifies calibration, and lowers overall cost. Additionally, the sensor’s low impedance output ensures stable readings and compatibility with a wide range of microcontroller ADC inputs without requiring complex buffering circuitry.

Physically, the LMT70 is a compact 0.88mm x 0.88mm footprint, making it one of the smallest analog temperature sensors available. This very small size enables integration into space-constrained designs. The device dissipates less than 36 µW, minimizing self-heating and preserving measurement accuracy even during continuous operation. With the characteristics of the LMT70, this temperature sensor is potentially a good selection for our RF energy harvesting sensor module.

3.5.3 Sensirion SHT41

The SHT41 is one of the latest low power sensor technologies developed by Sensirion. This sensing device includes two types of sensing measurements, temperature and humidity. Its operating current is approximately 0.4 µA, making it beneficial to use in our project for environmental sensing. The SHT41 performs best when within the recommended normal temperature and humidity range of 5 °C to 60 °C and 20% to 80% relative humidity (RH). Long term exposure to conditions outside the normal range may temporarily offset the signal of RH measurement (e.g. +3% RH after 60 hours at > 80% RH). Once returned to the recommended temperature and humidity range, the sensor will naturally recover within its specified performance limits. However, extended exposure to extreme conditions can accelerate device ageing.

A major benefit to using the SHT41 is that the sensor device supports I2C communication protocol. Multiple modes available include standard mode, fast mode, and fast mode plus. Data is transferred in multiples of 16-bit words. To increase reliability, I2C glitch protection is available on the SHT41 in the form of an 8-bit checksum.

To protect the sensor, the device comes with protective measures that you may choose from. The on-package filter membrane option for the SHT4x family provides an additional barrier for pollutants to enter the sensor opening, thus lowering negative influences on the sensing element. Mostly designed to keep particles and dust from accumulating and reducing the response time, the membrane also enables more efficient and easy cleaning, as it helps to reduce liquid intrusion into the sensor opening. The second option that is available to the SHT41 is it can be delivered with a protective cover attached such that the sensor opening is completely covered and sealed.

3.5.4 BME680

The BME680 is a small but powerful environmental sensor made by Bosch. It is designed to measure four things at once: temperature, humidity, air pressure, and gas levels. Many engineers use it in smart home devices, portable gadgets, and IoT systems that need reliable environmental data. Additionally, The BME680 is very small, only 3.0 x 3.0 x 0.93mm; it provides accurate readings while using very little power. It works over a wide range of conditions, including temperatures from -40°C to 85°C, and relative humidity from 0 to 100%. These ranges make it useful for indoor and outdoor projects, smart home devices, and portable electronics.

The BME680 communicates using digital serial interfaces, which makes it easy to connect to microcontrollers like STM32, Arduino, or Raspberry Pi. It supports both I2C and SPI, giving flexibility to project needs. In I2C mode, the sensor uses only two wires SDA for data and SCL for the clock making it ideal for low pin count designs or systems with many sensors on the same bus. The BME680 can operate at standard I2C speeds up to 3.4 MHz in high-speed mode. If a project needs faster data transfer or more stable communication in noisy environments, the sensor can switch to SPI mode, which uses four wires and supports clock speeds up to 10 MHz. Because it offers both protocols, the BME680 can fit into almost any embedded system without requiring special hardware.

For humidity measurement, the BME680 uses a high-quality capacitive sensor. It can measure humidity from 0% to 100% RH, covering the full range of real-world conditions. Its accuracy is ±3% RH, which is good for weather stations and environmental tracking. The sensor responds quickly, with a typical response time of 8 seconds, meaning it can detect changes in moisture levels very quickly. This is helpful in situations where humidity changes rapidly, such as near kitchens, bathrooms, or HVAC vents.

The temperature sensor inside the BME680 is designed to be both fast and stable. It typically achieves an accuracy of ±0.5°C, which is suitable for most consumer and educational applications. The sensor also has low noise and low drift, increasing its reliability over long periods without needing recalibration. Because the BME680 is so small, it can heat up slightly from nearby electronics, but this can be corrected in software applications by using compensation formulas provided by Bosch.

Like SHT41, this sensor can operate at ultra-low power, giving it an advantage when measuring temperature and humidity. At 1 Hz of humidity and temperature sensing, the sensor uses only about 2.1 µA, which is low compared to many other environmental sensors. This allows the device to operate on a PCB that will have ultra-low-power consumption.

3.5.5 HDC2080

The HDC2080 is a low power digital humidity and temperature sensor designed by Texas Instruments for HVAC systems, smart thermostats, and smart home assistants. It measures 0 - 100% relative humidity with a typical accuracy of ±2% RH, and temperature with a typical accuracy of ±0.2°C. The HDC2080 has a supply voltage range from 1.62 V to 3.6 V, making it compatible with low voltage microcontrollers and energy harvesting designs. The dimensions of the HDC2080 are 3.0 x 3.0 mm.

An advantage to the HDC2080 is its ultra-low power consumption. In sleep mode, it consumes only 50 nA, and at a sampling rate of one measurement per second, it consumes 300 nA for humidity readings only and 550 nA for a combined measurement of humidity and temperature. This makes it ideal for RF energy harvesting applications, where every microamp counts. The sensor also supports an auto measurement mode, allowing it to autonomously take readings at intervals without constantly intervening with the MCU, therefore reducing power.

The HDC2080 is also a capacitive-based sensor that includes integrated digital features and a heating element to dissipate condensation and moisture. This additional feature will help boost the performance of the sensor, allowing for more accurate data collection. This feature is especially useful in applications where humidity spikes could distort readings, applications that are prone to humidity spikes such as HVAC systems, outdoor enclosures, or rapidly changing indoor environments.

Digital features are another advantage of the HDC2080. It communicates over a standard I2C interface, supports programmable interrupt thresholds, and can trigger system wakeups when humidity or temperature crosses user defined limits. This reduces the need for continuous polling and helps microcontrollers remain in deep sleep modes longer. These capabilities make the sensor desirable for smart thermostats, home assistants, environmental monitors, and other IoT devices that require both accuracy and efficiency.

Overall, the HDC2080 is an accurate and power efficient environmental sensor that balances performance with extremely low energy usage. Its combination of high accuracy, integrated heater, autonomous measurement capability, and low power operation makes it a strong choice for modern embedded systems, especially those relying on RF energy harvesting.

3.5.6 Comparison and Selection of Environmental Sensors

Selecting the LMT70 and SHT41 together provides a strong balance between precision temperature measurement and reliable humidity sensing. The LMT70 excels as a dedicated analog temperature sensor, offering extremely high accuracy down to ±0.05°C in the human body temperature range and a wide operating span from -55°C to 150°C. Its small footprint and ultra-low power consumption makes it ideal for the energy harvesting sensor module, where thermal precision and power utilization is critical. Because the LMT70 outputs a linear analog voltage, it also integrates well with the STM32 Microcontrollers because of its ADC terminal.

The SHT41 will be strictly used for humidity measurement in this project, complementing the LMT70 by providing digital RH sensing with ±1% accuracy and extremely low energy consumption. While it also includes a temperature channel, relying on the LMT70 for temperature and the SHT41 solely for humidity ensures that each sensor is used for its strongest capability. Lastly, The SHT41 digital I2C interface simplifies humidity data acquisition and simplifies the complexity of our project. Together, the LMT70 and SHT41 form a robust sensing combination.

Specification	LMT70	SHT41	BME680	HDC2080
Sensor Type	Analog temperature sensor	Digital temp + humidity	Digital temp + humidity + gas + pressure	Digital temp + humidity

Interface	Analog voltage output	I2C	I2C	I2C
Supply Voltage	2.0 - 5.5 V	1.08 - 3.6 V	1.71 - 3.6 V	1.62 - 3.6 V
Current Consumption (While measuring)	12 µA (Max)	500 µA	340/350 µA (H/T)	650/550 µA (H/T) (typical)
Temperature Accuracy	±0.05°C (20-42°C)	±0.1°C	±0.5°C	±0.2°C
Temperature Range	-55°C to 150°C	-40°C to 125°C	-40°C to 85°C	-40°C to 85°C
Humidity Accuracy	N/A	±1% RH	±3% RH	±2% RH
Humidity Range	N/A	0 - 100% RH	0 - 100% RH	0 - 100% RH
Package Size	0.88 × 0.88mm	1.5 x 1.5 x 0.3 mm	3.0 x 3.0 x 0.93 mm	3.0 x 3.0mm
Special Features	Output enable pin, ultra-low power	Very low power, high accuracy	Multi-sensor environment	Low power, integrated heater

3.2 Electronic Display

3.2.1 Monochrome 1.3" 128x64 OLED Graphic Display

The 1.3-inch 128x64 monochrome OLED is a graphical display used in embedded systems that require visuals without the power from LCDs. Its OLED technology means each pixel emits its own light, eliminating the need for a backlight and producing extremely high contrast ratios. This makes the display readable in a wide range of lighting conditions, including complete darkness and bright indoor environments. The 1.3-inch display gives us a good balance between effective display and compactness in our transceiver design.

From an electrical standpoint, these displays typically use the SH1106 or SSD1306 drivers, both of which support I2C and SPI communication. This flexibility allows the display to integrate easily with our STM32 microcontrollers and other potential microcontroller units. The driver, integrated circuit handles low-level pixel control, memory addressing, and refresh operations, reducing the workload on the host MCU.

In terms of performance, the OLED panel offers quick response times that can reach under 10ms, making it suitable for animations and graphical data. Unlike LCDs, OLEDs maintain consistent contrast regardless of viewing angle, often reaching near 180° visibility. This makes the 1.3-inch module ideal for handheld devices and compact instrumentation where the user may view the screen from different angles. The display brightness is also adjustable through software, allowing developers to optimize power consumption or improve visibility depending on the application.

Power efficiency is one of the strongest advantages of this display. Because OLED pixels only consume power when lit, screens with mostly black backgrounds draw significantly less current. A typical 1.3-inch OLED consumes around 20 - 30 mA at moderate brightness, which is far lower than comparable LCD modules with LED backlights. This makes the display well-suited for battery powered devices such as compact user interfaces. Developers can further reduce power usage by dimming the display, updating it less frequently, or using sleep modes built into the device drivers.

Mechanically, the module is lightweight and typically mounted on a small PCB with a 7-pin or 4-pin header. The screen itself is thin and durable, with no glass backlight layer, making it more resistant to shock and vibration than LCDs. Overall, the 1.3-inch 128×64 OLED is a versatile, high-contrast, low-power display that fits a wide range of embedded applications requiring clear graphical output.

3.2.2 Monochrome 0.96" 128x64 OLED Graphic Display

The 0.96-inch 128x64 OLED is a display used in embedded systems due to its compact size, low cost, and excellent visual clarity. Despite being smaller than the 1.3-inch version, it retains the same 128x64 resolution, resulting in a higher pixel density and sharper graphics. This makes it ideal for small devices where space is limited, but a readable graphical interface is still required. The display's monochrome nature simplifies rendering and reduces memory requirements on the microcontroller.

Most 0.96-inch OLED modules use the SSD1306 driver, which has become a standard in the embedded community. The SSD1306 supports both I2C and SPI, though many modules are wired for I2C by default to minimize pin usage. The driver includes a full display buffer, contrast control, and multiple addressing modes. Because of its popularity, nearly every major microcontroller ecosystem, Arduino, STM32, ESP32, and Raspberry Pi has mature libraries that make the display very easy to use.

Visually, the display offers excellent contrast and readability thanks to OLED technology. However, the small size does limit the amount of information that can be displayed at once, but clever UI design such as using icons, scrolling text, or multi-page menus can overcome this limitation. The display is also highly readable from wide angles, making it suitable for wearable devices, compact controllers, and measurement tools.

Power consumption is very low, typically around 10 - 20 mA depending on brightness and pixel usage. Because the display is so small, even full brightness operation is efficient compared to larger OLED or LCD modules. This makes the 0.96-inch OLED a favorite for battery powered IoT devices, sensor nodes, and handheld gadgets. The SSD1306 also includes sleep modes that can reduce power draw to microamp levels when the display is not in use, further extending battery life.

Mechanically, the module is extremely compact and lightweight. It usually comes with a 4-pin header for I2C communication. The PCB is small enough to fit into tight enclosures, and the display itself is thin and durable. Many developers choose this module for prototypes because it is inexpensive, easy to mount, and supported by a large ecosystem of libraries and tutorials. Overall, the 0.96-inch 128x64 OLED is a reliable, efficient, and highly accessible display for small embedded systems.

3.2.3 Elegoo 0.96 Inch OLED Display Screen

Another electronic display is the Elegoo 0.96-inch OLED module built with an SSD1306 driver. With a resolution of 128x64 pixels, it is useful for displaying sensor data and simple animations while remaining extremely power efficient. Because it uses OLED technology rather than a backlit LCD, each pixel emits its own light, resulting in sharp contrast and excellent readability even in lowlight environments. This makes it ideal for battery powered embedded systems and handheld devices such as our user interface and power transmitter device.

One strength of the Elegoo OLED module, is its communication interface. This electronic display supports I2C communication. I2C is a convenient communication tool for our project because our project will be consistent with mostly using I2C. Regardless of the interface type, the SSD1306 protocol is well documented and widely supported across microcontroller systems, including the STM32 MCU family.

The display operates from 3.3 V to 5 V, proving it to be compatible with the STM32WB series. It's typical operating current is around 12 - 25 mA depending on pixel usage, making it suitable for sensor designs. Because OLED pixels only consume power when lit, designs that use mostly black backgrounds can significantly reduce energy consumption. This characteristic is especially valuable in IoT applications.

From a usability standpoint, the Elegoo OLED is straightforward to integrate. Libraries from the STM32 HAL based drivers provide functions for drawing text, shapes, and other elements. The displays small footprint also makes it easy to mount on PCBs, where it is intentional for or handheld transceiver device to be small.

Overall, the Elegoo 0.96-inch OLED display is a reliable module that fits well into the embedded systems side of our project. Its combination of low power consumption, I2C communication, and library support and accessible instructions make it a good choice to use for displaying the data we need.

3.2.4 SunFounder I2C LCD1602 Display Module

Lastly, the SunFounder I2C LCD1602 module is a classic LCD designed to display two lines of text, each containing 16 characters. Unlike graphical OLEDs, this display is ideal for projects that require readable text without the complexity of pixel-level graphics. Also, it is supported by almost every microcontroller platform.

Like the other electronic displays discussed, the SunFounder LCD has similar communication protocol, I2C. This dramatically simplifies the integration necessary with our STM microcontroller. The I2C interface also allows multiple sensors and peripherals to share the same bus, making it ideal for designs that are limited to its pin availability. Additionally, from a programming perspective, the SunFounder LCD1602 is easy to use. Libraries exist for multiple IDE's including STM32 HAL based drivers and various Python modules for Raspberry Pi. These libraries provide simple functions necessary for utilizing this LCD.

The LCD1602 operates using a backlit liquid crystal display, which provides good readability in both bright and dim environments. The backlight is usually LED-based and can be controlled or dimmed through software. However, the power consumption is modest usually around 15 - 20 mA with the backlight on, making it usable for portable devices, though not as energy efficient as OLED modules.

In practical embedded systems, the LCD1602 excels in applications where text output is sufficient, such as sensor readouts, and device status messages. Its robustness, simplicity, and wide compatibility make it a staple in educational kits, prototyping environments, and industrial control panels. For STM32 projects that require straightforward text output without the overhead of graphical rendering, the SunFounder I2C LCD1602 remains a reliable and accessible choice.

3.2.5 Comparison and Selection of Electronic Display

Selecting the Elegoo 0.96-inch OLED is the best electronic display for this RF energy harvesting project. Because it uses an OLED panel rather than an LCD, every pixel emits its own light, giving high contrast and readability in low light environments. The 128x64 resolution is more than enough for sensor readouts, icons, and simple graphics. Therefore, making it a strong choice for handheld devices, portable modules, and battery powered projects. Its small footprint also pairs well with the idea of being easy to use, allowing you to integrate a display without significantly increasing the device's size or power budget.

From an engineering standpoint, the display's I2C interface simplifies wiring and firmware development, especially when working with our STM32 microcontroller(s). The SSD1306 driver is widely supported across libraries, which reduces development time and ensures stable performance. The OLED's low current consumption helps maintain long battery life. Overall, the Elegoo 0.96-inch OLED offers a dependable solution to display environmental sensing data, while simplifying the complexity of the system.

Electronic Display	**Monochrome 1.3" 128x64 OLED Graphic Display**	**Monochrome 0.96" 128x64 OLED Graphic Display**	**Elego 0.96 Inch OLED Display Screen**	**SunFounder I2C LCD1602 Display Module**
Size	5.6 x 33 x 6.2mm	29.2 x 26.7mm	27.0 x 27.0 x 4.0 mm	124.46 x 43.18 x 10.16mm.
Display Type	OLED	OLED	OLED	LCD
Resolution	128x64	128x64	128x64	16x2
Supply Voltage	1.6 V - 3.3 V	1.6 V - 3.3 V	3.3 V - 5 V	5 V
Operating Temperature	-40°C to 85°C	-40°C to 85°C	-20°C to 60°C	N/A
Driver	SSD1306 and GFX	SSD1306 driver	SSD1306 driver	STM32 HAL driver library
Interface	I2C/SPI	I2C	I2C	I2C
Cost	$19.95	$17.50	$9.99	$9.99

3.3 - Power Harvesting IC

The core of this project is its ability to passively harvest energy from a 915MHz ISM band signal. This will be captured by a 915MHz antenna and rectified by a voltage-doubling full-wave rectifier. As such, one of the most important components to consider is the power harvesting IC. There are 2 major aspects of performance to consider. The first is start-up power. Start-up power is the minimum amount of available power required for the power harvesting IC to begin collecting energy. This parameter directly relates to the maximum range that the sensor node can be activated from, as power is the direct quantity that dissipates as distance increases. The second parameter is efficiency. Efficiency in this context is the amount of power that can be stored as usable energy in the reservoir capacitor vs the total amount of power that has been captured by the antenna. This parameter is directly related to charging time of the circuit, or the amount of time that the sensor node must receive power for until it has harvested enough energy to complete a sensing cycle.

Harvesting IC	Start-Up power (V, uW)	Conversion Eff. Max (%)	Comments
BQ25570	0.6, 15	90	16s MPPT check
LTC3108	0.02 (2Ω), 200	50	E for Vin = 50mV
P2110B	-12dBm = 63	60	50Ω Z-matched Very expensive
NEH7110	0.27, 12	95	LDO output
AEM30330	0.27, 3	90	Lowest Power CS Manual Z-match

A power harvesting IC comparison table would be incomplete without the inclusion of the Texas Instruments BQ25570. It was one of the premier energy-harvesting integrated circuits for a very long time. It had incredible efficiency, integrated both a buck converter and a boost converter, was built to explicitly work with battery storage set-ups, and most importantly integrated a maximum power point tracking logic system. With a microscopic 15uW start-up power, nearly any reasonable power source at the time would be able to easily turn the IC on and begin harvesting. One of its main downsides, however, is that it requires 600mV of potential to begin harvesting. The datasheet does list a minimum harvesting voltage of 100mV. However, this is only the case when the circuit is already powered on. Because of the fact that our circuit will be in a hard shut down mode for most of the time they are deployed, the true turn on voltage for these harvesting systems is the cold-start voltage. In this case, where our diodes only have reverse voltage ratings of typically up to 4V, a 600mV minimum turn-on voltage for the system severely limits the maximum amount of power that our system can absorb. Additionally, its maximum power point tracking system is hard-set to a 16 second testing period. For this application, where charging time is an important metric of performance, 16 seconds of delay between applying the RF power source and the sensor node being able to efficiently absorb it is far too long of a time. It is for these reasons that the BQ25570 was not chosen for this application.

The Analog Devices LTC3108 is a legendary power-harvesting integrated circuit for one reason: it has a minimum cold-start voltage of 20mV. For applications such as this one where the amount of voltage received by the harvesting antenna will be on the order of millivolts on start-up, this may sound like an incredibly attractive option. 20mV on a 50Ω source is equivalent to a sensitivity of –21 dBm (8uW). This, however, is a misleading line of reasoning. At 20mV, the LTC3108 requires an input resistance of 2Ω. The true power required by the 20mV LTC3108 is therefore (20mV)^2)/2 = 200uW, equating to only –7dBm. Furthermore, the conversion efficiency of this IC maxes out at only 50%. Even the Texas Instruments BQ25570 reaches over 90% conversion efficiency. A monumental 200uW start-up power at 50% efficiency is already quite literally orders of magnitude worse than even the Texas Instruments BQ25570, which itself is middle of the pack. Furthermore, due to the RF nature of this project, we have the ability to impedance match our circuit with a high-Q matching filter and boost the voltage received by our antenna to match even such an IC as the BQ25570 with its huge 600mV start-up potential. It is for this list of reasons that the LTC3108 was not chosen for this application.

At first glance, the PowerCast P2110B seems like a solution to our project without even having to do any work. It is a 50Ω matched power harvesting IC designed specifically to absorb 915MHz ISM-band radiated power from an antenna all the way down to –12dBm and store it in either a battery or a capacitor for use in a passive device. Fortunately for the integrity of our senior design project, this is not the case. -12dBm correlates to a start-up power of 63uW. This is over four times worse than the Texas Instruments BQ25570 and over twenty times worse than the lowest start-up energy chip researched. Even if one takes into account the power losses associated with the rectification diodes that one must use in the DC-based harvesting chips, the PowerCast P2110B falls short. Additionally, its power conversion efficiency is laughable. Sporting an absolute maximum power conversion efficiency of only 60% at 0 dBm and an absolutely pitiful 8% at its marketed minimum operating power of –12 dBm, the only chip that the PowerCast P2110B beats is the Analog Devices LTC3108. Furthermore, putting aside any pure engineering requirement, the PowerCast P2110B has a purchasing price of over $50 USD. This astronomical cost immediately places it out of the single-component budget for this project and out of the budget for any cheap IOT device that this integrated circuit would be best put to use inside of. Any integrated circuit on this list, perhaps excluding the Analog Devices LTC3108, would outperform this chip given the proper matching network and maximum power point tracking setup. It is for these reasons that the PowerCast P2110B was not chosen for this application.

The Nexperia NEH7110 is one of the two relatively modern power harvesting integrated circuits on the research list. A cold start potential of 275mV at only 15uW immediately puts it ahead of nearly every other chip on the list. It beats the Texas Instruments BQ25570 in terms of both power and cold start potential and beats both the Analog Devices LTC3108 and PowerCast P2110B massively in terms of power. What really sets this chip apart from the rest is its conversion efficiency. The NEH7110 boasts a cold-start conversion efficiency of over 75%. This already puts it ahead of both the PowerCast P2110B and the Analog Devices LTC3108. Furthermore, it has a maximum power conversion efficiency of 95%, better than any other researched power harvesting integrated circuit. Interestingly, the Nexperia NEH7110 opts to integrate a low-dropout regulator for its output conversion rather than a more directly efficient switching converter. This is likely done to minimize its on-state current consumption while the rest of the circuit is in a sleep mode. In an energy harvesting circuit where sleep-mode quiescent currents for the entire circuit are measured in nanoamps, the amount of energy required to keep the switching converter running could be larger than the rest of the circuit combined. A low-dropout regulator does not have this switching loss and its power consumption is directly proportional to the power that it is providing to the rest of the circuit. For this application, where the circuit will be in a hard shut-down mode rather than sleep, quiescent current is of less concern and direct conversion efficiency is a more important metric to consider. Furthermore, an LDO can only be used to buck voltage down. A capacitive energy storage element where stored energy is directly related to the voltage across it would thus be limited in its energy storage to the maximum voltage the harvesting IC can push across it minus the minimum operating voltage of the circuit. In some cases, that could cut the available energy of the circuit in half or more, effectively reducing the conversion efficiency below 50%. The Nexperia NEH7110 is undoubtedly a strong contender for this application, but it does have flaws.

The e-peas AEM30330 takes the cake for minimum power consumption. Despite having a similar minimum cold-start potential to the Nexperia NEH7110 of 275mV, it requires only one quarter of the power, starting up with only 3uW (–25 dBm) of input, smaller than any other researched power harvesting IC by an astounding four times. This number alone could be the difference between our one-meter basic goal and our 3-meter advanced goal. While it does not reach the astronomical 75% start-up efficiency of the NEH7110, it is still no slouch. At 275mV of input, it is able to convert 65% of the input power to usable energy. Furthermore, it integrates a buck-boost converter that offers high energy conversion in both storage and usage. The maximum power point tracking function of the AEM30330 is programmable to between 18ms and 72s which represents the widest range of MPPT testing periods. For this project, we will likely use the 1 second MPPT period. If MPPT is not desired and a static input resistance more accurately describes the source, the AEM30330 has the option of manually defining the input resistance of the onboard DC-DC converter to guarantee maximum power transfer. The AEM30330 also beats the NEH7110 in quiescent current (850nA vs 1500nA) despite integrating a switching converter. Quiescent current does not present a particularly large gain in engineering value as this application will spend the majority of its time in a hard shut-off mode, however, it is beneficial to note that this circuit is generally more efficient as a whole compared to the NEH7110. Overall, the AEM30330 satisfies the widest range of engineering requirements while simultaneously requiring few, if any, compromises. It is for these reasons that the AEM30330 was chosen for this application.

3.4 - Energy Storage Element

3.4.1 - Batteries vs Capacitors

Micro-power applications are one of the few areas where the decision between batteries and capacitors must be made. At the microjoule level, the energy density of capacitors becomes less of an issue than it is normally. Because of this, their many advantages, often soundly offset by this single downside, begin to shine. Therefore, a research section comparing the benefits of battery storage with the benefits of capacitor storage is warranted. We begin with the following table.

	Batteries	Capacitors
Leakage	Low	Low
Voltage vs Energy	Constant	Changing
Energy Density	High	Low
Life Cycle	Moderate	High
Charge / discharge rate	Moderate	High
ESR	Low	Low

Though this application will likely be required to charge the full amount of energy required for a sensing cycle each time it is pinged, leakage current is still an important factor to consider. The circuit will be charged over the course of multiple seconds at an input current of microamps. Therefore, a leakage current of even a few hundred nanoamps could severely lengthen the amount of time that is required for the circuit to charge. It would be misleading to say that either batteries or capacitors have lower leakage currents. Certain styles of batteries have more and less leakage current compared to certain styles of capacitors, and certain styles of capacitors have less leakage currents than certain styles of batteries. For example, solid-state batteries have lower leakage currents than any supercapacitor and film capacitors have lower leakage currents than any lithium-ion battery. It is therefore incorrect to state that capacitors or batteries have lower leakage currents compared to batteries or capacitors; both technologies include designs that outcompete the other. The lowest leakage current capacitors are those of the film type, and the lowest leakage current batteries are those of the solid-state type. We will therefore investigate the qualities of film capacitors versus solid state batteries for this application.

The AEM30330 integrates a buck-boost converter that can be used to convert effectively any voltage from the storage element into the required operating voltage for the circuit. The disadvantage of capacitors losing voltage as their stored energy is consumed is therefore a non-issue. In this application, the direct voltage-energy relation of capacitors may, in fact, be a large advantage. This circuit will only be consuming on the order of 100uJ of energy under the worst-case scenario (10mA for 10ms). For a solid-state battery, determining a difference of 100uJ is a practically impossible task under the constraints of this application. In order to determine when the circuit has gathered enough energy to complete a full sensing cycle we must therefore include a separate circuit to determine the amount of energy absorbed by the system and alert the AEM30330 when the threshold is reached. This increases both the complexity and the cost of the overall system. The energy in a capacitor is directly related to the voltage across it, and the AEM 30330 is directly designed to sense this voltage and begin supplying power when it reaches the desired threshold. For this reason alone, capacitors may be the superior choice for our energy storage device.

The single biggest downside to capacitive energy storage is its overall low energy density. Even the most modern high-density supercapacitors top out at around 24 Watt-Hours per Kilogram [x]. This is compared to a standard, cheap lithium-ion battery, which averages about 200 W*h/Kg [x]. This is exaggerated by the fact that supercapacitors generally come with many downsides for this application. For example, their equivalent series resistance is notably high for capacitors. This translates to lower discharge efficiency compared to other types of capacitors. Their self-discharge rate can be on the order of microamps [x]. This application will likely be gathering on the order of microamps of current. A supercapacitor would discharge faster than the harvesting IC could fill it. Fortunately, this application will not be storing a significant quantity of energy. At worst case of 10mA for 10ms, this sensor circuit will consume 100uJ of energy. The AEM30330 has a minimum storage capacitance of 100uF. Only a few film capacitors are required to satisfy this requirement, and only 1.5V must be stored across them to reach the 100uJ energy requirement. Most film capacitors are capable of safely storing tens to hundreds of volts, and the AEM30330 is capable of storing up to 5V across the energy storage device. This corresponds to over 1mJ of energy, far more than this project requires. The low energy density of capacitors is therefore a non-issue for this application. A film capacitor energy storage device at 100uF will easily be able to store over 10x the energy required for this circuit to even complete 1 cycle.

Lifecycle is a particularly important parameter to discuss in this application. This sensing node will be placed in difficult to reach environmental areas where temperature control is not possible, and replacement periods even on the order of decades become a hassle. Lithium-ion batteries tend to have cycle lifetimes on the order of thousands of charge-discharge cycles. While this sounds like plenty of cycles, this disregards the effect of temperature. Batteries are fundamentally chemical components, and temperature has a direct effect on their operation. As temperature increases, the efficiency, lifetime, capacity, etc., of batteries decrease. Non-chemical film and ceramic capacitors, whose energy storage is squarely defined by the purely electromagnetic $E = (C*V^2)/2$, are effectively impervious to these environmental effects. Furthermore, non-chemical capacitors have an effectively infinite lifetime. The chemical membranes within batteries and chemical-capacitors degrade every charge-discharge cycle. The dielectric membranes within film capacitors and the ceramic dielectric of ceramic capacitors do not. Capacitors have lifetimes that exceed those of the PCB itself.

Charge / discharge rate and ESR are closely related concepts when discussing batteries and capacitors. Batteries, whose electrical attributes are defined by the chemistry that powers them, are constrained in their charge / discharge rate by the speed of their chemical reaction. For typical lithium ion batteries, their “C-Rate”, or the ratio of their discharge rate to capacity, is typically around 1. For a standard 1-C lipo battery with a capacity of 1000mAh, the maximum current it can discharge is about 1A. A 2.2uF polypropylene film capacitor can discharge around 6.5A (dc) [x]. Even at the maximum capacity of 0.5*(2.2uF)*450^2 = 0.22J = 15uAh at 3.7V, this corresponds to a C rate of 6.5/15e-6 = 433,000. Lithium-ion batteries typically have equivalent series resistances on the order of milliohms to tens of milliohms [x-2]. Film capacitors offer similar ESR. The difference, however, is that for energy storage, many film capacitors are typically placed in parallel. When placed in parallel, the ESR is also put in parallel, and the overall ESR is therefore greatly reduced. ESR in the hundreds of microohms is not uncommon for highly parallel capacitor energy banks.

After analyzing the pros and cons of batteries versus capacitors, two things stand out in favor of capacitors. Firstly, they have effectively unlimited lifetimes. In a set-and-forget application such as this, replacing batteries every decade is unfeasible. Secondly, they display their contained energy directly with the voltage placed across them without any external circuitry required. It is for these reasons that capacitive energy storage was chosen for this application.

3.4.2 - Capacitor Technologies

The simple capacitor has an astoundingly diverse range of implementation methods. The digikey "capacitors" category alone describes Ceramic Capacitors, Aluminum Electrolytic Capacitors, Film Capacitors, Aluminum - Polymer Capacitors, Tantalum Capacitors, Electric Double Layer Capacitors aka Supercapacitors, Tantalum Polymer Capacitors, Thin Film Capacitors, Mica Capacitors, and PTFE Capacitors. The following sections and tables will discuss the differences between each type, their strengths, and their weaknesses. The maximum capacitance available to purchase is found by opening the digikey website, specifying the capacitor type, and sorting by capacitance descending. Tan δ, or dissipation factor, is the product of capacitance, signal angular velocity, and equivalent series resistance. It measures how lossy a capacitor is. Energy density is derived by taking the maximum capacitance value found earlier and multiplying by either 5V or the capacitors maximum rated voltage, whichever is lower. 5V was chosen as a stopping point because it is the maximum voltage that the AEM30330 can induce across its energy storage element. Leakage / insulation resistance describes how long a capacitor can hold a voltage. Ageing describes how much of a capacitors original capacitance is lost over time, typically measured in percent per decade-hour, or percent / log10(hours). When comparing a technology, we must first define what is important for the application. The most important specifications for this application are, in order of importance, the leakage current, ageing, and dissipation factor of the capacitors that we are investigating. Maximum capacitance and energy density in this case are very tightly coupled as the maximum voltage allowed across the capacitor is only 5V, far below the rated voltage for nearly every capacitor on this list. Because of the relatively small 100 microfarad capacitance required for this circuit, these two specifications are less important than the rest. Leakage is by far the most important because it directly affects the charge speed and thus the maximum reasonable range of the circuit. Even a leakage current of a few microamps represents a power loss greater than the total input power available to not only the circuit but to the harvesting antenna itself. It is therefore crucial that the chosen capacitor has a leakage current preferably close to or less than single nanoamps. Ageing is the second most important specification. These devices will be placed in difficult to access areas that best make use of their wireless capabilities. It is therefore integral to the project that these devices need not be replaced on even 5 to 10 year timescales. One of the major reasons that capacitors were chosen over batteries was for the effective lack of aging that certain capacitor types exhibit. Capacitors that age are therefore suboptimal for this application. The third most important specification is dissipation factor, which relates to the equivalent series resistance of the capacitor. When discharging, the current in all capacitors flows through this equivalent series resistance. This represents both a voltage drop and more importantly power loss. In this application, any and all lengths must be taken to reduce power loss and increase efficiency. Thankfully, the resistance of capacitors is typically measured in the range of milliohms, so this becomes less of an issue in a circuit that is only flowing at absolute most 10mA of current. These three specifications are what we will use to compare the available capacitor implementation technologies. The capacitor type that best embodies them will be chosen for use in this application.

	C max Farads	Tan δ - 1/Q Resistance	E density Joule / cm^3	Leakage/Ins. Res.	Ageing % / dBh
Ceramic					
X5R	470u	7.5-20% .	0.206 .	500 MΩ*uF	5%
X7R	200u	2.5-10 .	0.006 .	500 MΩ*uF	3% .
C0G/NP0	2u	0.1-0.5% .	0.00012 .	1 GΩ*uF	0%
Film					
Polyester	560u	1%	0.07	3 GΩ*uF .	<1.5% .
Polypropylene	270u	0.1-0.2%	0.034	30 GΩ*uF .	0%
Electrolytic	2.2	8-20% .	0.02 .	3 uA, <200K .	5k hr
ELDC/SC	13.5k	3-250mΩ	32.4 .	8uA - 1mA	1500 hr

Ceramic capacitors are a natural starting point for any capacitor technology comparison. Small in size, relatively large in capacitance, and built for high-frequency operation, the ceramic capacitor sits right next to the transistor when it comes to most important modern inventions. Ceramic capacitors are differentiated from each other in purpose by the dielectric used to form them. There are many different ceramic dielectrics, so for the sake of simplicity, only the three most common have been added to the comparison list. We begin with the X5R dielectric. Of the three, it sports the highest relative permittivity and can therefore be used to create the highest capacitance capacitors of the three dielectrics and therefore in this case the highest energy density. Its leakage current is relatively low, equal to the X7R dielectric of 500 MΩ*uF which at 100uF corresponds to a resistance of 5 MΩ. At 1V, this represents a leakage current of 200nA, barely within our range of acceptable leakage currents. Where it falls short is in its ageing. Ceramic capacitors undergo capacitance ageing, where capacitance is lost as time goes on. This ageing factor is presented in terms of percent per decade-hour, or percent capacitance lost as time increases logarithmically. The X5R dielectric undergoes on average 5% capacitance degradation every decade. At 10,000 hours (roughly a year) this capacitor would lose 20% of its overall capacitance. This is unacceptable for our application. Fortunately, other dielectric materials exist. Closest to X5R in performance is the X7R dielectric. Its leakage current is the same, and its maximum capacitance is about half that of the X5R dielectric, but it also drops its ageing factor almost in half, only losing about 3% capacitance per decade-hour. This still corresponds to 12% over just one year, which is also suboptimal for this design. Losing capacitance means losing energy. The same voltage that may have been able to power the circuit a year ago may no longer be able to, and this kind of loss is explicitly unacceptable, and for these reasons, the X5R and X7R dielectric ceramic capacitors will not be selected for usage in this design. The third dielectric, C0G/NP0, seemingly fixes these problems. It exhibits a capacitance degradation of zero percent and doubles the insulation resistance of both X5R and X7R. Furthermore, its dissipation factor is at least over 25x less than X7R and at least over 75x less than X5R, representing a significantly lower equivalent series resistance and therefore greater circuit efficiency. C0G/NP0 capacitors unfortunately come with a major downside. Their maximum capacitance is typically measured in tens to hundreds of nanofarads. Even the largest C0G/NP0 capacitor available on digikey is only 2uF, and available capacitors do not even reach 1uF, capping out at about 0.94. This application would require over 100 of these capacitors in parallel to reach the 100uF minimum value required by the AEM30330. This is impractical and expensive. For these reasons, ceramic capacitors will not be chosen for this application.

The aluminum electrolytic capacitor was a standard in the home of any hobby electronics enthusiast. If a circuit needed a capacitor before the 1990s, that circuit used an aluminum electrolytic capacitor. Their high capacitance, manufacturing and purchasing affordability, convenient packaging, and acceptable (at the time) frequency response meant that in any home, an aluminum electrolytic capacitor would not be far. Near-ubiquity then, for better or worse, does not mean a competitive product now. Even then, their lifespans were almost a joke in any circle where they were used. Blowing a cap was so widespread that it even made its way into common American slang! This is corroborated in most electrolytic capacitor datasheets available. A measly 5k hour lifespan (given, at their maximum rated voltage) corresponds to only half a year! This is made exponentially worse by their well-known temperature sensitivity. If this application is to be used in hard to reach areas, it should be expected that many of these are in an unprotected natural environment. In Florida for example, where ambient temperatures regularly exceed 100 degrees Fahrenheit, and enclosed areas bombarded by sunlight can even exceed 140, it would be a death sentence for any electronics engineer to choose aluminum electrolytic capacitors. This is compounded by their high dissipation factor of 8-20%, not even beating out the X5R ceramic capacitor in terms of efficiency. The final nail in the coffin for aluminum electrolytic capacitors is their horrific microamp-scale leakage currents. In situations where they are not used for energy storage, or of such storage is on the order of joules to kilojoules of energy, such leakage currents may be acceptable. In this microjoule-scale application, leaking even 1uA of current could mean cutting our range by a factor of 10. This is an unacceptable amount of loss. It is for these reasons that aluminum electrolytic capacitors will not be chosen for this application.

Electric Double Layer Capacitors, more widely known as supercaps / supercapacitors, are something of a celebrity in the world of capacitors. As far as power goes, they hold most of the advantages of capacitors without the major disadvantage of poor energy density, reaching even up into the range of some types of batteries. As far as energy density goes, ELDC Supercaps are thousands of times better than even the closest contender. This is, unfortunately, their only advantage for this application. In the realm of ESR, they rank at the bottom, capping out in the hundreds of milliohms. Their lifetimes and temperature sensitivity are even worse than that of aluminum electrolytic capacitors. Because they rely on the phenomenon of pseudocapacitance, a chemical effect, the same thermal effects and time-degradation of any chemical system also affect them. It is because of this that their lifetimes are measured in thousands of hours, an unacceptably low duration. The single worst aspect for this application, however, is their leakage currents. 8uA minimums mean that even at the bottom end, ranges over 1m are unachievable. Furthermore, charging duration vastly increases as the same current that was just used to charge the battery is lost to ground 80% as fast as it was received. Supercapacitors are therefore an abysmal choice for this application, and it is for these reasons that they will not be chosen for this application.

Film capacitors are known for their exceptionally high breakdown voltages. Typical aluminum electrolytic capacitors rarely exceed operating voltages of 10V, and electric double layer supercapacitors rarely exceed operating voltages of 5V. Ceramic capacitors are slightly better, with operating voltages of 50V not uncommon. The lowest rated operating voltage radial film capacitors available on digikey bottom out at 50VDC. Typical operating voltage maximums extend well into the hundreds of volts, and certain film capacitors are even rated for thousands of volts. While, as stated earlier, maximum voltage rating is not necessarily an important aspect of this project, such high voltage ratings do reveal something very important about these capacitors. For such high voltage ratings, the dielectric used in these capacitors must be exceptional. Their more important aspects, such as dissipation factor and dielectric resistance, corroborate with this expectation. Polyester film capacitors demonstrate dielectric resistances of around 3 gigaohm-microfarads. This is 3x better than the best ceramic capacitors and is orders of magnitude more than the best aluminum electrolytic / electric double layer supercapacitors. Polypropylene film capacitors are even better, exhibiting around 30 gigahom-microfarads of insulation resistance. No other kind of capacitor even approaches this level of leakage. At 100uF, we will be leaking only 3pA of current at 1V. At this level, flux leakage from residual after soldering can become more of an issue if the PCB is not cleaned properly. Furthermore, their dissipation factors are some of the lowest available in any capacitor. Polyester capacitors exhibit typical loss tangents of around 1%, corresponding to incredibly low equivalent series resistances. There are only two capacitors that beat this. The first is the C0G/NP0 ceramic capacitor, which is hugely limited by its tiny maximum capacitance. The second is the polypropylene film capacitor. Polypropylene film capacitors can exhibit dissipation factors in the 0.1-0.2% range. This is superior to every other capacitor researched, bar none. Additionally, in the case of polypropylene film capacitors, the issue of ageing is not present, because they do not age. There is only one other capacitor on this list that does not age, that being the C0G/NP0 ceramic. There are, however, some tradeoffs between the two dielectrics. While polypropylene has superior insulation resistance, loss tangent, and does not exhibit ageing, polyester capacitors have a higher capacitance density.

In this case, the advantages of polypropylene still vastly outweigh the capacitance issue; it is possible to find single polypropylene capacitors with capacitance values up to and over 100 microfarads. Polypropylene film capacitors shine in all three of the major categories that this application cares about. They do not age, exhibit near-lossless dielectrics, and effectively do not leak. It is for these reasons that polypropylene film capacitors will be chosen for this application.

3.4.3 - Comparison of Film Capacitors

3.5 RF Power Transfer

3.5.1 Choosing a Frequency Band

Sending power to the energy harvesting sensor board will be done using an RF signal. To ensure the project is compliant with FCC regulations, the transmitted RF signal must be in an unrestricted frequency band and use an allowable amount of power (See Chapter 4.1 for more information). The most used frequency bands for hobbyists and prototype projects are the 902 MHz to 928 MHz and 2.4000 GHz to 2.4835 GHz bands, which are set aside by the FCC for Industrial, Medical, and Scientific (ISM) purposes. Among these two options, the 902 MHz to 928 MHz band (typically referred to by its center frequency as the 915 MHz band) is the better option for wireless power transfer due to propagation loss.

The free space path loss (FSPL) of a signal traveling in free space, or a vacuum. Vacuum permittivity, denoted by ε_0, is a constant used to describe how electric fields interact and propagate through mediums. Other mediums besides vacuums can be described using the permittivity constant, such as how air is equal to $1.0006\varepsilon_0$. Due to how close the permittivity of vacuum and air are, they will be considered near equal when determining the FSPL of the transmitting RF signal. FSPL can be calculated as part of the Friis transmission equation, which describes power transferred from one antenna to another in vacuum, and is shown below:

$$P_r^{[dB]} = P_t^{[dB]} + G_t^{[dBi]} + G_r^{[dBi]} + 20\log_{10}\left(\frac{\lambda}{4\pi R}\right) \quad (3.1)$$

In the equation, Pr is the power transmitted to the receiving antenna, and Gr is its gain. Similarly, Pt is the power of the transmitting antenna, and Gt is its gain. The term in parenthesis models the FSPL where the loss increases linearly with signal frequency, which includes wavelength (λ) and the distance between transmitting and receiving antennas (R). To compare the 915MHz band to 2.4GHz, we only look at the FSPL portion of the Friis equation since the other terms will not matter when keeping the antennas in both cases the same. Wavelength can be determined by dividing the speed of light by the signal frequency, and from there we can determine the FSPL. Given the basic goal of power transmission at

1 meter of separation, we find that the FSPL at 915MHz and 2.4GHz is –31.7dB and –40.1dB, respectively. As such, choosing a lower frequency is preferable for the power transmission in our project.

3.5.2 Single Frequency versus Spread Frequency

The amount of power which can be used to transmit the signal is limited depending on the type of transmission used. Per FCC guidelines, a radio transmitter using a single carrier frequency in the 915 MHz ISM band is allowed for a maximum radiated power of –1.23 dBm in a given direction. This radiated power is the combination of the transmitter power, transmitter gain, and any losses that occur between them. To provide usable power to our receiver board and sensors, the RF signal must be rectified into a DC current that can be stored as charge in a battery or capacitor. Looking back at equation 3.1, a maximum allowable radiated power of –1.23 dBm is not enough power to offset attenuation from FSPL without using an incredibly high-gain receiving antenna on the order of 30dBi or higher. A gain this large is not reasonable to design or buy without using additional amplification after the receiver, which will not work due to the receiving board being entirely passive.

The FCC allows for higher radiated power when using spread spectrum transmission techniques, which use a quickly changing carrier frequency to move the signal around the band. Many consumer radio frequency integrated circuits support the use of Frequency Hopping Spread Spectrum (FHSS) – a type of spread spectrum technique – which in turn allows for a radiated power of up to +30 dBm when using less than 50 hopping frequencies and +36 dBm when using more than 50. This is a thousand-fold increase over the maximum power for a singular carrier frequency, making it highly desirable for our wireless power transmission application. However, spreading the power transmission across the 915 MHz band means the frequency of the RF signal can vary from 902 MHz to 928 MHz. This relates to a relatively small fractional bandwidth at only 2.8%. However, the receiving antenna must be designed to have high efficiency in the entire bandwidth instead of only 915 MHz. This is a slight downside to using FHSS for power transmission, but the massive increase in allowable power makes it a worthwhile tradeoff.

	Single Frequency	FHSS
Allowable Power	-1.23 dBm	+30 dBm for < 50 channels +36 dBm for > 50 channels
Transmitter Integration	More complex	Less Complex
Receiver Complexity	Lower	Higher
Cost	Lower	Higher

When comparing design characteristics of signal frequency and FHSS, the number of pros and cons are tied. Due to 915 MHz being a common transmission band for radio communications, consumer chips that include FHSS are commonly available. In contrast, transmitting a single frequency without regard to modulation or mixing of data transmission requires at least an oscillator and amplifier. Consumer chips also have the added benefit of datasheets with design notes on chip integration along with recommended part values, making them far easier to implement in our design. The cost of an RFIC would be higher than using discrete components; however, their supplier's documentation saves design time and prevents errors which helps lower costs elsewhere.

3.5.3 RFIC Packages

3.5.3.1 Analog Devices ADF7020BCPZ

The ADF7020BCPZ is a high performance, low power ISM band transceiver IC that supports both 433 MHz, 868 MHz, and 915 MHz transmissions. It is a highly integrated chip capable of supporting FSK, ASK, and OOK which makes it highly desirable for low power radio communication devices. The chip comes in a compact 48-pin VFQFN package with 0.5mm pitches, making it a reasonable choice for compact board layouts while still being able to be hand soldered. The device is suitable for commercial use and adheres to the regulations outlined in the North American FCC Chapter 15 and supports a multitude of modulation schemes for transmitted and received signals. In addition, the ADF7020BCPZ includes a VCO and low noise fractional-N Phase Locked Loop (PLL) that allows it to be used in frequency hopping systems, allowing the device to have a higher allowable output power. The maximum output power of the chip is +13 dBm, which is far below the maximum allowed value of +36 dBm for systems with many hopping channels, but the system gain can be increased by including an amplifier in the RF front end. When transmitting at +10 dBm, which is near its maximum output power, the ADF7020BCPZ draws a current of 26.8 mA.

A key design choice that sets the ADF7020 apart from other RFICs is its low intermediate frequency receiver architecture. While most receivers will down convert the signal to the baseband, this architecture instead converts to an intermediate frequency before digital demodulation. This reduces the receiver’s susceptibility to DC offsets, flicker noise, and self-mixing effects that tend to hurt devices that directly down convert their signals. Furthermore, the ADF7020 accomplishes this without using many additional external components, which helps with design integration. Another aspect of the ADF7020 which sets it apart is its VCO that operates at twice the desired RF frequency. This feature helps to reduce spurious out of band emissions in restricted frequency bands, which help to ensure the chip adheres to regulations while delivering solid performance.

The ADF7020BCPZ can use either a 3-wire programming interface or SPI depending on the microcontroller being used. Analog devices provide driver libraries for interfacing with the RF transceiver for two of the microcontrollers – the ADuC841 and the ADSP-BF533 – and the connection scheme will differ depending on the microcontroller which is used. Communication with a microcontroller gives the design the option to place the IC into low-power mode, in which it only consumes 0.1 µA, giving the chip a highly efficient standby mode option.

3.5.3.2 Semtech SX1262

Semtech's SX1262 is a long range low power transceiver with a maximum output power of +22dBm that supports transmissions in the 915 MHz ISM band. The active current consumption during transmission is 45 mA at +14 dBm and climbs to 118 mA at +22 dBm. Inside the chip is a protocol engine which manages any modulations or protocol schemes the designer wishes to use. The IC achieves this high output power level with its highly-efficient integrated front end power amplifier, which reduces the need for an external amplifier. The SX1262's low power consumption makes it an effective component for use in IoT applications, telemetry, smart metering, building automation, and environmental sensing. The SX1262 is also a highly integrated RFIC, as it contains a digital communication block, analog and data conversion front end, LoRa and FSK modulator-demodulators (modems), and internal power regulation. This drastically reduces the number of external components required to integrate the RFIC with the rest of the hardware.

Transmitting as much power as possible is the primary goal when using the RFIC. As mentioned before, the use FHSS will be necessary to achieve the maximum allowed power for our 915 MHz transmitter. The SX1262 supports both FHSS and conventional narrowband modulation schemes. In Long Range (LoRa) mode, the programmable bandwidth range is 7.8 kHz to 500 kHz and the spreading factor – which is how long an individual transmission remains on air – is selectable as well. Narrow bandwidths and higher spreading factors result in slower transmission speeds since signals are more likely to interfere with other devices. With this in mind, the transmission rate of the SX1262 ranges from 0.018 kbps when using the highest spreading factor and narrowest bandwidth and can go as high as 62.5 kbps when using small spreading factors and large transmission bandwidths. In addition to LoRa, the SX1262 supports Long Range Frequency Hopping Spread Spectrum (LR-FHSS) which is a modulation scheme developed by Semtech to improve network capacity in already communication networks where long range wireless communication is necessary.

Programming and communication with the SX1262 is done peripherally through a host microcontroller using an SPI interface. While the SX1262 has internal registers which determine operational characteristics like transmission gain, bandwidth, and modulation scheme, the host microcontroller can communicate with the RFIC using a command-based API. The drivers for the API are supplied by Semtech directly and are open source on GitHub, which makes software integration straightforward for the SX1262. Interfacing over SPI requires seven wires for this chip, which is three more than typical. SPI requires a SCLK line, MOSI, MISO, and the optional chip select if multiple devices are on the communication bus. In addition to these, there is a BUSY line, a general purpose data input/outputs (DIOs), and an SX_NRESET line used for factory reset. The BUSY line is controlled by the SX1262 and tells the host microcontroller when it is ready to receive a new command. The line will go low only when ready for a new command and is kept high at all other times. Per the datasheet, at least one DIO is required for interrupt (IRQ) handling.

These interrupts are used by the SX1262 to indicate when transmission is done, receiving is done, either of these timeout, or when an error occurs. SPI communication with the SX1262 is synchronous full-duplex, meaning data can be sent both ways simultaneously.

3.5.3.3 Texas Instruments CC1101

The CC1101 by Texas Instruments is a low-power transceiver for use in bands below 1 GHz, including 915 MHz. This chip has excellent documentation and is very inexpensive, making is a popular and widely used choice for both professional and hobbyist RF communication designs. A microcontroller can communicate with the CC1101 over SPI, with drivers and example code being available from TI as part of the component's documentation. Reference design boards with component values are also easily obtained from TI's product support page which significantly aides in PCB design for our project.

The supported frequency bands include 300-348 MHz, 387-464 MHz, and 799-928 MHz, and within these bands the CC1101 supports 2-FSK, 4-FSK, GFSK, ASK, and OOK. For wireless power transfer, we do not care as much about the modulation scheme used since only power is being transmitted. However, the large array of options would give us flexibility to choose a modulation scheme that can transmit a higher average power. The maximum transmission rate is 600kbps at a maximum power of +12 dBm. The CC1101 can operate at voltages between 1.8 V and 3.3 V, making it easy to integrate alongside other peripherals and the microcontroller that all typically use 3.3 V. Typical current consumption is 14.7 mA in receive mode and 34 mA in transmit mode when using maximum power. When in power-down mode, or sleep mode, the current draw is as low as 0.2 μA which makes the CC1101 a good choice for battery-powered platforms. However, while it is low the current draw is larger than that of the SX1262 which sits at 0.16 μA.

The CC1101 is programmed using 4-wire SPI from a host microcontroller. Texas Instruments provides a driver library which makes software development easier for the designer. However, getting the most out of the RFIC transmitter is not as straightforward as calling a function to set the output power. Instead, the CC1101 uses internal registers to set its operational characteristics and Texas Instruments directs the designer to use their SMARTRF-STUDIO software to help evaluate transceiver performance and determine the best register settings for the desired operation. Furthermore, while the CC1101's datasheet is very informative and in-depth, its coverage of software development focuses on low level registers and complex timings. TI's drivers should do most of the heavy lifting when it comes to low level timing and register settings, however if anything is to be added to the functionality of the device then we would need to interact with the chip's complex architecture.

Texas Instruments also makes CC1190, which is a sister chip for the CC1101 that acts as an RF front end to amplify the output of the transmitter. When following TI's recommended board layout and component selection, this will increase the system output power to +27 dBm which is far greater than most affordable commercial RFICs. The SX1262 can also achieve this gain, and even exceed it, if an RF front end power amplifier is used alongside it. However, since the CC1190 and CC1101 are both made by Texas Instruments there is a multitude of application notes and verified board designs which we can reference to ensure our PCB can reach +27 dBm output power. The CC1190 contains a high-power amplifier, low-noise amplifier, impedance matching circuitry, and control logic. The chip also interfaces with the host microcontroller using SPI, which helps consolidate our communication topologies.

3.5.3 Comparison of RFIC Transceivers

Between the three RFICs researched, the Semtech SX1262 provides the best all-around solution for our project's wireless power transmission due to its high transmit power, software support, and innate implementation of LoRa and FHSS transmission schemes. Although the Texas Instruments CC1101 can achieve up to +27 dBm when paired with the CC1190 and the ADF7020 contains a highly capable RF front end, neither RFIC provides the same performance and ease of implementation as the SX1262. Furthermore, the SX1262 can be made to achieve as much, if not more, transmit power as the CC1101 and CC1190 combination when also using an RF front end power amplifier, of which the CC1190 is. Below is a comparison table summarizing the key characteristics of each RFIC, with standout figures and features of each chip highlighted in green.

	ADF7020	SX1262	CC1101
Frequency Range	431-478 MHz, 862-956 MHz	169-315 MHz, 433-490 MHz, 868-928 MHz	300-348 MHz, 387-464 MHz, 799-928 MHz
Max Data Rate	200 kbps	62.5 kbps	600 kbps
Communication Protocols	2-FSK, 3-FSK, 4-FSK, ASK, OOK	LoRa, LR-FHSS	2-FSK, 4-FSK, GFSK, ASK, and OOK
Supports FHSS	No	Yes	Yes
Max TX Power	+13 dBm	+22 dBm	+12 dBm
TX Current	26.8 mA	118 mA	34.2 mA
Sleep Current	< 1 µA	0.16 µA	0.2 µA
Driver Support	For Analog Devices µCs	Widely Available from both supplier and users	Available from Supplier
Hardware Complexity	Medium	Medium; reference design available but additional hardware is recommended	Low; TI gives reference designs
Software Complexity	Medium; requires an Analog Devices µC	Low; API-based and many examples online	Medium; register based with helpful TI software

3.6 Antennas

3.6.1 Monopole Antenna

Monopoles are one of the simplest and most widely used types of antennas, consisting of only one conductor and a ground plane. The conductor is mounted above the conductive ground plane, which acts as an electrical mirror to form the equivalent of a dipole antenna. The typical length of a monopole antenna is one-quarter wavelength ($\lambda/4$) of the frequency which will be used, and their short length makes them preferable for sized-constrained applications.

The gain of a monopole antenna depends on the geometry and design of its ground plane. A monopole with an infinitely large ground plane can achieve a theoretical gain of around +2.15 dBi. In real designs where the PCB is relatively small and its components will exhibit losses and quarter wavelength monopole antennas will typically achieve only +1.5 to +2.3 dBi. Increasing the length of the monopole antenna causes an increase in its gain, with larger variants normally being 5/8 wavelengths and achieving +3 to +5 dBi gain. The gain of a monopole antenna is created by compressing the radiated RF energy horizontally in a toroidal shape. This toroid extends all around the antenna, with the majority of energy being focused perpendicular to the body of the antenna. This gives monopole antennas relatively low directivity compared to other antennas which focus their radiated energy into narrower beamwidths.

Given that monopole antennas are widely used across many applications, the complexity of integration into our system is straightforward. They are vertically polarized, meaning the electric field is parallel to the radiating antenna. Polarization is an important factor to consider when choosing antennas for our system, since optimal power transfer requires both transmitting and receiving antennas to use the same type of polarization. Monopole antennas are most often fed with a 50 Ω transmission line and require only a simple LC matching network.

Typical applications of monopole antennas include wireless sensor networks, RFID readers, and remote-control devices, where all of which require omnidirectional communication. Depending on the size requirements of the given application, the monopole antenna can be implemented in a multitude of ways, including: a straight wire antenna; a whip antenna, which is a monopole made of flexible material; a helical antenna, made compact by winding it about its center; as a PCB trace antenna on rigid or flexible material.

3.6.2 Dipole Antenna

A dipole antenna is a type of antenna which uses two conductive elements of equal length to transmit and receive RF signals. The length of these elements should each be one-quarter of the signal wavelength used with the antenna, making the total antenna length equal to half the wavelength of the signal. The antenna is connected to the circuit at its center where the two halves meet, called the feedpoint, which also acts as a virtual ground. This can simplify circuit implementation since a separate ground plane is not required for antenna operation. The typical gain for a half-wavelength dipole antenna is 1 – 2 dBi, which is relatively low compared to other antennas. Increasing the length of the dipole to five-fourths the desired resonant wavelength can improve the gain to around 5 dBi, but in turn requires a significant increase in antenna footprint.

3.6.3 Yagi Uda Antenna

A Yagi Uda is a highly directional antenna which is constructed from multiple parallel conductive elements that focus the radiated RF energy into a narrow beamwidth. One of these conductive elements is the driven element, which is the only one in the array that is directly connected to the transmitter or receiver. The driven element is typically half wavelength in size and is a dipole antenna. Behind the driven element is the reflector, which is designed to be slightly longer than the driven element by around 5%. This increase in size compared to the driven element causes its impedance to be parasitically inductive, thus any RF energy absorbed by the reflector will create a current that lags behind the current on the driven element. A lagging current creates destructive interference with the driven element, hence any radiated energy in the direction of the reflector will be significantly less than that in the transmitting direction of the Yagi Uda antenna. In front of the driven elements are additional conductors of smaller size called directors. Similarly to the reflector, the shorter length causes it impedance to be capacitive when compared to the driven element, thus its current will lead ahead of its voltage and create a constructive interference in the direction of transmission. The constructive interference is what gives Yagi Uda antennas high gains and directionality, and increasing the number of director elements further increases both characteristics.

Simple Yagi Uda antennas have 3 elements (a driven element with a reflector and one director), which gives them a directional gain of +5 to +8 dBi. Increasing the number of elements gives nonlinear increases in gain, where 5 driving elements has around +10 dBi and 8 elements has +13 dBi [x]. Furthermore, since the elements in Yagi Uda antennas are all roughly 1/2 wavelength in size and each element is separated by 0.2 to 0.25 wavelengths, the overall size of the antenna will be larger than dipole and monopole antennas. For instance, the wavelength at 915 MHz is 32.8 cm which means a 3-element antenna will be around 0.75 wavelengths in length, or 24.6 cm. Compared to the monopole and dipole antennas, which will be 8.2 cm and 16.4 cm respectively, the total size and weight of the Yagi Uda make integration into the handheld transceiver more complex.

3.6.4 Comparison of Antennas

Criteria	Monopole	Dipole	Yagi Uda
Size	$\lambda/4$	$\lambda/2$	Multiple wavelengths, $\geq 2\lambda$
Typical Gain	2 – 5 dBi	1 – 2 dBi	7 – 15 dBi

Directivity	Low	Low	High
Beamwidth	Wide	Wide	Narrow
Bandwidth	Broad	Broad	Moderate
Polarization	Linear	Linear	Linear
Cost	Low	Low	High

3.7 Communication Protocols

3.7.1 Hardware Communication

In this section, we examine the various communication protocols available to use for our handheld transceiver and our sensor node module. The communication protocols are used to communicate between the MCU, sensors, transmitters, and receivers. This is a critical decision for the project because it affects the complexity and the power consumption of our low-power sensor module. This choice also depends on the capabilities and requirements of the other hardware on the devices.

3.7.1.1 I2C

I2C offers a compact and modular communication interface, relying on only two lines SDA for data and SCL for the clock, which reduces pin count and simplifies hardware design. Because all devices share the same bus, adding new peripherals requires no additional routing beyond assigning a unique address, making I2C valuable for sensor systems. It supports multi-master and multi-slave configurations and is well suited for low-speed, low-power peripherals commonly found in embedded and IoT applications. However, its bus architecture introduces limitations, these limitations include restricted bus speed and cable length, pull-up resistors potentially increase power consumption if not optimized, and the protocol itself is more complex than SPI or UART due to start/stop conditions and signaling. Considering our selected microcontroller for this project, this communication method is our preferred choice for easy event management and for our temperature/humidity sensors control.

3.7.1.2 UART

UART provides an asynchronous communication method that requires no clock line, making timing less rigid and configuration straightforward. One feature of the STM32 microcontroller for our sensor board is that it contains an LPUART peripheral specifically optimized for ultra-low-power operation, which is advantageous for battery powered or energy harvested systems. Features such as half duplex single wire mode and modern operations allow flexible communication capabilities. However, some limitations stem from its design: each UART channel supports only two devices, preventing separate configurations without additional hardware. It also requires accurate baud-rate matching to avoid framing errors, making it less ideal for systems with many peripherals.

3.7.1.3 SPI

SPI excels in applications requiring high data rates, supporting clock speeds in the tens of megahertz, and enabling full duplex communication. Its dedicated clock line ensures deterministic timing, and its signaling makes it ideal for high-speed sensors, displays, and memory devices. However, SPI's speed comes at the cost of increased pin usage, each slave device typically requires its own chip select line, complicating PCB routing and higher power consumption compared to protocols like I2C and UART. SPI lacks built-in error checking mechanisms. This means that additional software or hardware measures are required to ensure the integrity of data, increasing complexity in error-prone environments.

Specification	**I2C (Inter-Integrated Circuit)**	**SPI (Serial Peripheral Interface)**	**UART (Universal Asynchronous Receiver/Transmitter)**
Communication Type	Synchronous, multi-master/multi-slave	Synchronous, full-duplex	Asynchronous, point-to-point
Number of Wires	2 wires (SDA, SCL)	4+ wires (MOSI, MISO, SCLK, SS)	2 wires (TX, RX)
Topology	Multi-device bus	Master-slave	One-to-one
Clocking	Clock provided by master	Clock provided by master	No clock signal (asynchronous)
Complexity	Moderate (addressing, protocol overhead)	Higher (multiple lines, manual chip-select)	Low (simple byte-based communication)
Best Use Cases	Sensors, low-speed peripherals, shared bus systems	High-speed data transfer, displays, ADCs/DACs	Serial communication, debugging, long-distance, low-speed links

3.7.2 Wireless Communication

In this section, we evaluate the wireless communication technologies available for transmitting data from our environmental sensing board to the handheld transceiver and any supporting network infrastructure. Wireless communication is a core design element of the system because it determines how reliably sensor measurements can be delivered, how far the data can travel, and how much power the sensing module must consume during operation. Since the environmental sensing board is responsible for collecting and transmitting environmental data, the choice of wireless protocol directly affects battery life, update frequency, and overall system responsiveness.

Selecting the appropriate wireless technology also requires balancing range, throughput, interference tolerance, and network topology. Each protocol, whether long-range LoRa, high bandwidth WiFi, low energy BLE, or mesh capable Zigbee offers distinct advantages and constraints that influence how the sensing board interacts with the handheld transceiver. These considerations ensure that the final design supports reliable communication while meeting the project's low power and environmental monitoring requirements.

3.7.2.1 LoRa

LoRa is a low power, long-range wireless communication technology designed for IoT applications that require devices to transmit small amounts of data over several kilometers. Operating in sub-GHz ISM bands (typically 433 MHz, 868 MHz, or 915 MHz), LoRa uses Chirp Spread Spectrum (CSS) modulation to resist interference and reliable communication in both rural and urban environments. Its ability to penetrate obstacles and maintain connectivity makes it ideal for sensor networks, smart agriculture, environmental monitoring, and industrial automation.

A major strength of LoRa is its extremely low energy consumption, allowing battery powered nodes to operate for years without maintenance. This long-range capability comes at the cost of low data rates, typically between 0.3 kbps and 11 kbps. LoRaWAN, is a software application built with the integration of LoRa modulation, which defines how devices use the LoRa hardware. Despite its limited throughput, LoRa remains one of the most effective technologies for long-range IoT systems.

LoRa communication is organized into several functional subcategories that define how devices transmit, receive, and manage data within a long-range, low power network. These subcategories include Device Classes (A, B, and C), Spreading Factors, Adaptive Data Rate (ADR), and Regional Parameters. Together, these elements determine communication timing, energy usage, signal robustness, and regulatory compliance across different geographic regions.

Device Classes define how often a LoRa device listens for downlink messages.

- Class A devices are the most energy-efficient, opening two short receive windows only after transmitting an uplink message.

- Class B devices add scheduled receive windows using periodic beacons, enabling more predictable downlink communication.

- Class C devices keep their receive window open nearly continuously, providing low latency downlink at the cost of higher power consumption.

LoRa also uses Spreading Factors (SF7 - SF12) to balance range and data rate. Higher spreading factors increase range and robustness but reduce throughput, while lower spreading factors provide faster data rates with shorter range. The Adaptive Data Rate (ADR) mechanism automatically adjusts spreading factors and transmission power based on network conditions, therefore improving efficiency and extending battery life. Finally, Regional Parameters define frequency plans, duty cycle limits, and channel configurations for different geographic areas, ensuring compliance with local regulations while maintaining interoperability across global deployments.

3.7.2.2 WiFi (IEEE 802.11)

WiFi is a high throughput wireless communication standard often used for networking in homes, offices, and industrial environments. Operating primarily in the 2.4 GHz and 5 GHz ISM bands and more recently, the 6 GHz band with WiFi 6E, WiFi provides significantly higher data rates than other short-range wireless technologies. Modern WiFi standards such as 802.11ac (WiFi 5) and 802.11ax (WiFi 6) support speeds exceeding 1 Gbps under ideal conditions, making WiFi suitable for applications requiring large data transfers, continuous connectivity, or integration with cloud-based systems. For our environmental sensing board, WiFi offers a robust pathway for transmitting sensor data to a handheld transceiver, a local gateway, or a remote server when high bandwidth or network infrastructure is available.

One of WiFi's major advantages is its ability to support complex, multidevice networks with high reliability and strong security features. WiFi networks use advanced modulation schemes to maintain stable communication even in environments with heavy interference or many connected devices. This makes WiFi a strong candidate for scenarios where the sensing board must operate in dense RF environments or communicate with multiple devices simultaneously. Additionally, WiFi's widespread adoption ensures compatibility with smartphones, laptops, embedded modules, and IoT gateways, simplifying integration and user interaction.

However, WiFi's high performance comes with tradeoffs that must be considered for low-power sensor systems. WiFi modules typically consume more energy than BLE, Zigbee, or LoRa, especially during active transmission or when maintaining continuous connectivity. This can significantly impact battery life if the environmental sensing board is expected to operate for extended periods without recharging. To mitigate this, modern WiFi chipsets support low power modes, short wake cycles, and optimized transmission intervals, allowing devices to send bursts of sensor data and quickly return to sleep. Despite these limitations, WiFi remains an excellent option when the sensing board requires direct internet access or integration with existing network infrastructure.

3.7.2.3 BLE Bluetooth Low Energy

Bluetooth Low Energy (BLE) is a short-range wireless communication protocol designed specifically for ultra-low-power operation while maintaining reliable connectivity between embedded devices. Operating in the 2.4 GHz ISM band, BLE uses frequency hopping spread spectrum (FHSS) across 40 channels to minimize interference and maintain stable communication in environments with heavy RF congestion. BLE is widely used in wearables, medical sensors, and smart home devices because it enables small battery powered nodes to operate for months. For our environmental sensing board, BLE offers a lightweight and energy efficient method for transmitting sensor data to nearby devices such as the handheld transceiver.

One of BLE's defining strengths is its flexible communication model. BLE supports both connection and broadcast oriented modes, allowing devices to either maintain a persistent link or simply advertise data periodically without forming a full connection. This makes BLE ideal for applications where the sensing board only needs to send small packets of environmental data at regular intervals. BLE's data rate up to 2 Mbps with Bluetooth 5 provides more than enough throughput for sensor measurements while keeping energy consumption extremely low. Additionally, BLE's low latency and fast connection setup allow the sensing board to wake up, transmit data, and return to sleep quickly, preserving energy.

BLE also includes several advanced features that enhance performance in IoT systems. Bluetooth 5 Long Range can extend communication distances up to 100 meters or more by using lower data rates and stronger error correction, making BLE viable even in moderately spaced sensor deployments. Bluetooth Mesh, built on top of BLE, enables many communications across large networks, allowing sensor nodes to relay messages through intermediate devices. Although mesh networking increases complexity and power usage, it provides an option for distributed sensing systems. BLE's widespread support across smartphones, tablets, and embedded modules ensures excellent compatibility, making it a practical choice for diagnostics, configurations, and data retrieval from the environmental sensing board.

3.9.2.4 Zigbee (IEEE 802.15.4)

Zigbee is a low power, low-data-rate wireless communication protocol designed for mesh networking in IoT and automation systems. Operating primarily in the 2.4 GHz ISM band though regional variants exist at 868 MHz and 915 MHz. Zigbee supports data rates up to 250 kbps, making it suitable for sensor applications that transmit small, periodic packets of information. Its communication stack is built on the IEEE 802.15.4 Physical/MAC layer, which provides reliable short-range wireless links with minimal energy consumption. This efficiency allows Zigbee devices to operate long-term on small batteries, a key advantage for distributed sensing systems.

A defining feature of Zigbee is its mesh networking capability, which enables devices to forward messages on behalf of others. This architecture dramatically increases network resilience and coverage: if one node fails or becomes unreachable, data can automatically reroute through alternate paths. Zigbee networks can scale to hundreds of nodes, supporting dynamic joining and leaving without disrupting overall communication. These characteristics make Zigbee particularly effective in environments where many devices must coordinate, such as smart home installations, large commercial buildings, and industrial automation systems.

Zigbee's practical strengths extend beyond its technical architecture. It is widely adopted in HVAC control, security systems, and environmental monitoring platforms. Although its range is shorter than long-range protocols like LoRa and its data rate is modest compared to Wi-Fi, Zigbee strikes an ideal balance for dense, low-power IoT deployments where reliability, scalability, and energy efficiency matter more than raw throughput. For applications such as environmental sensing where nodes periodically report temperature, humidity, air quality, or motion, Zigbee provides a robust and flexible wireless backbone that maintains consistent communication across a distributed network.

3.7.2.5 Comparison and Selection of Wireless Communications

Communication	LoRa	WiFi	BLE	Zigbee
Frequency Band	433/868/915 MHz	2.4/5/6 GHz	2.4 GHz	2.4 GHz (also 868/915 MHz variants)
Typical Range	2 - 15km	30 - 50m	10 - 100m	10 - 100m (mesh extends coverage)

Data Rate	0.3 - 11 kbps	11 Mbps - 46 Gbps	Up to 2 Mbps	Up to 250 kbps
Power Consumption	Very Low	High	Very Low	Low
Best Use Cases	Long-range IoT, agriculture, environmental sensing	Internet access, high-bandwidth devices	Wearables, mobile peripherals, Smart tech	Smart home, automation, sensor networks

3.8 Software

The applications of software and hardware are equally important in many engineering projects. The importance of choosing the right software for our projects will define how our devices will operate and their ability to collect data accurately. Research into our software applications will also determine how our devices will communicate with each other. Our project is based on two transceivers that will be highly dependent on software applications that allow effective communication.

We will first describe and compare software applications used to design a Printed Circuit Board (PCB). Continuing from this, we will compare the types of Integrated Development Environments (IDE's) that could be used in programming our microcontrollers.

3.8.1 PCB Design Software Applications

3.8.1.1 KiCad

KiCad is one of many tools used for PCB design and is especially popular because it is completely free. KiCad provides a collection of tools for schematic capture, PCB layout, footprint creation, and 3D visualization, making it a capable solution for electronics design. For this project, KiCad offers all the essential features needed to create schematics, define board layouts, and verify component placement before fabrication. Its 3D viewer is useful for visualizing the final PCB assembly, checking mechanical clearances, and ensuring that sensors and components align properly with enclosures.

A major benefit of KiCad is its ability to import PCB designs and libraries from many other software applications, including Fusion 360, Altium Designer, and SolidWorks. This flexibility makes it easier to collaborate with teams using different tools or to migrate designs into a modern environment. KiCad also includes a built-in SPICE simulation engine, offering functionality like LTSpice. It supports four standard simulation modes AC sweep, DC transfer, operating point, and transient analysis allowing users to validate circuit behavior before committing to a PCB layout.

Because KiCad is an opensource platform, users have access to a large network of shared libraries and tutorials. The software is updated frequently, with new features and performance improvements by both volunteer contributors and industry partners. KiCad also avoids the licensing restrictions found in many commercial tools, allowing unlimited board size, unlimited layers, and unrestricted use for commercial projects.

However, KiCad does have limitations. While its feature set has grown significantly in recent years, it still lacks some of the advanced features found in premium software like Altium Designer. The user interface, while improving, can feel less intuitive for beginners, and certain workflows such as differential pair tuning or complex rule-driven routing may require a more manual setup. Additionally, because KiCad relies heavily on community-maintained libraries, users need to verify footprints and symbols to ensure accuracy.

3.8.1.2 Fusion 360

Another software application that is available for PCB design is Fusion 360, which is useful for the construction of electrical schematics and PCB layout. A benefit to using Fusion 360 is that it allows you to have an all-in-one workflow to design electronics. Using fusion 360 gives the user more flexibility for easily going back to earlier stage sketches or dimensions without having to rebuild your entire model. Lastly, fusion 360 is useful in allowing multiple users to have access to working on the same project.

Some cons of this software application are that it is more expensive than other software applications. Fusion 360 is also known to be dependent upon its cloud services, which can be frustrating if you lose internet connection and all the work you have done. Fusion 360 has also had performance and crashing issues due to complex schematics or layouts.

A further consideration when evaluating Fusion 360 for PCB design is its learning curve. Because Fusion 360 integrates mechanical CAD, simulation, and electronics design into a single environment, new users may need time to become comfortable navigating its broader toolset and understanding how different workspaces interact. Additionally, while the unified workflow is powerful, it can feel overwhelming for teams that only need lightweight schematic capture or PCB layout without the added mechanical features. Despite these challenges, Fusion 360 remains a strong option for multidisciplinary projects, especially when electrical and mechanical teams need tight integration and shared access to evolving design files.

3.8.1.3 Altium Designer

Lastly, another commonly used software application for PCB design is Altium Designer, which provides a comprehensive environment for schematic capture, PCB layout, and advanced electronics design workflows. One of the major benefits of using Altium Designer is its integrated toolchain schematics, PCB layout, simulation, and documentation are all connected within a unified interface. This integration reduces design headaches and helps maintain consistency across all stages of development. Altium also offers powerful routing tools, and advanced features such as interactive length tuning, making it a good choice for high-speed or complex multilayer boards.

Another advantage is its strong collaboration ecosystem. Altium 365 allows teams to share projects, review designs, and manage version control through a cloud-based platform. This makes it easier for multiple engineers to work on the same design, track changes, and maintain a clean revision history. The software also integrates well with mechanical CAD tools through STEP export/import, enabling collaboration between engineering sections.

Some cons to Altium Designer, similar to Fusion 360, are its costs and cloud services. The software also has a steeper learning curve compared to more lightweight PCB tools, requiring time to fully understand its advanced features and workflow. While Altium 365 provides strong cloud collaboration, it also introduces some dependency on online services, which may be inconvenient in environments with limited connectivity. Lastly, Altium can experience performance slowdowns when handling extremely large or highly detailed designs, especially on lower-end hardware.

3.10.1.4 Comparison and Selection of PCB Design Software

KiCad is the preferred design platform for this project because it offers a powerful, fully open-source workflow without required licensing associated with tools like Altium Designer or Fusion 360. Its schematic and PCB layout tools are intuitive and useful for developing our environmental sensor module and handheld transceiver device. KiCad also provides extensive libraries and integration with 3D modeling, without using expensive subscriptions. This makes it a flexible choice that aligns with the project’s emphasis on accessibility and modeling of our project's electronics.

3.8.2 Integrated Development Environments

3.8.2.1 STM32CubeIDE

STM32CubeIDE is the main software used for programming STM32 microcontrollers, and it is designed to make working with these chips much easier. One advantage is the built-in

CubeMX tool, which lets you set up the microcontroller visually instead of writing long configuration code. You can choose which pins to use, how fast the system clock should run, and which peripherals you want to enable. This saves a lot of time and helps prevent mistakes, especially when dealing with complex hardware features. The IDE also includes a full compiler and debugger, so you can write code, build it, and test it all in one place.

Another strength of STM32CubeIDE is how well it integrates with ST's official libraries. These libraries provide functions for controlling timers, ADCs, communication interfaces, and more. Because everything is designed by the same company that makes the microcontroller, the tools work together smoothly and reliably. The IDE also supports advanced features like Bluetooth Low Energy, USB communication, and real-time operating systems, which makes it suitable for more advanced embedded projects. This makes STM32CubeIDE a strong choice for our team project.

However, STM32CubeIDE can feel overwhelming because it includes many features and settings. It also runs heavier than simpler tools, so it may feel slow on older computers. Another limitation is that it is mainly designed for STM32 devices, so it is not as flexible if you want to work with different microcontroller brands. Compared to the Arduino IDE, it is much more powerful but also more complex. Compared to Visual Studio Code, it is less customizable but more specialized. Overall, STM32CubeIDE is the best option when working specifically with STM32 microcontrollers and when you need strong debugging and hardware control.

3.8.2.2 Visual Studio Code

Visual Studio Code (VS Code) is customizable code editor that becomes extremely powerful when you include extensions. By itself, it is simply a clean and fast environment for writing code, but with tools like PlatformIO, CMake Tools, or Cortex-Debug, it can support many different microcontroller platforms. This flexibility makes VS Code a popular choice for developers who want one environment that works for STM32, Arduino, ESP32, and many other boards. Because it is not tied to any specific company, it gives you the freedom to build your own workflow.

One advantage of Visual Studio Code is how well it supports project organization and collaboration. It includes built-in Git tools, a terminal, and smart code suggestions that help you write and debug code more efficiently. PlatformIO, one of the most popular extensions, provides a unified way to manage libraries, compile code, and upload firmware across many different microcontrollers. This makes Visual Studio Code especially useful for larger projects or teams that need a flexible and modern development environment.

The downside is that Visual Studio Code does not include built-in tools for configuring microcontroller hardware. For STM32 development, you still need to use STM32CubeMX or manually write initialization code. Debugging also requires extra setup, which can be confusing for starters. Compared to STM32CubeIDE, Visual Studio Code gives you more freedom but requires more effort to configure. Compared to the Arduino IDE, it is far more powerful but not as simple. Visual Studio Code is best for developers who want a customizable environment and are comfortable setting up their own tools.

3.8.2.3 Arduino IDE

The Arduino IDE is designed to be simple and easy to learn, which is why it is often the first tool to use when starting with microcontrollers. The interface is clean, and most tasks such as selecting a board, choosing a port, and uploading code only take a few clicks. The Arduino libraries hide many of the complicated hardware details, so you can focus on writing code that makes LEDs blink, sensors read values, or motors move. This makes the Arduino IDE perfect for quick experiments, classroom projects, and early prototypes.

Another major benefit of the Arduino IDE is its large library ecosystem. Almost every sensor, module, or device has an Arduino library that makes it easy to use without needing to understand low-level hardware registers. The IDE also includes a serial monitor and plotter, which help you see data from your microcontroller in real time. Because the Arduino community is so large, there are countless tutorials, examples, and forums that make troubleshooting much easier. This support is one of the reasons the Arduino IDE remains very popular.

However, the Arduino IDE has limitations when compared to more advanced tools. It does not include professional debugging features, and it does not give you much control over the microcontroller's internal settings. This makes it less suitable for projects that require precise timing, low power operation, or advanced communication protocols. Compared to STM32CubeIDE, it is much easier to use but far less powerful. Compared to VS Code, it is simpler but not nearly as flexible. The Arduino IDE is best for beginners and rapid prototyping, but not for complex embedded systems.

3.8.2.4 Comparison and Selection of IDE

STM32CubeIDE was selected as our programming environment because it provides a fully integrated development for the specific microcontrollers we will be utilizing in this project. Its integration with STM32CubeMX allows us to configure clocks, peripherals, and other components. Additionally, it allows us to generate initialization code that aligns perfectly with ST's HAL and LL libraries. This reduces setup time, minimizes configuration errors, and ensures our firmware is built on officially supported drivers. The IDE also includes a powerful debugger with real-time variable monitoring and compatibility with ST-Link programmers, making it ideal for iterative development on STM32 devices. By using STM32CubeIDE, we gain a stable environment optimized specifically for the STM32WB05KZ and the broader STM32 ecosystem, improving reliability and development efficiency.

IDE	Supported Coding Languages	Notes
STM32CubeIDE	C, C++	Best for STM32 hardware; low-level embedded programming.
Visual Studio Code	C, C++, Python, and many more	Depends on extensions; most flexible and customizable.
Arduino IDE	Simplified C/C++ (Arduino language)	Easiest to use; great for beginners and quick prototypes.

4.0 Related Standards and Design Constraints

4.1 FCC Regulations for Intentional Radiators

Use of radio frequencies, whether for communication or wireless energy transfer, falls under the regulation of the Federal Communications Commission (FCC). For the purposes of this project, the main regulations to be concerned with are the maximum allowed power that can be radiated from our transceiver and the frequencies bands which are available for unrestricted use. The regulations for radiated power are listed in Part 15 of Title 47 from the Code of Federal Regulations issued by the FCC. Section 15.209 lists the maximum allowable field strength for intentional and spurious emissions. Together, Sections 15.209 and 15.205 establish the regulatory framework by which unlicensed intentional radiators must abide.

Frequency (MHz)	Field Strength (µV/m)	Measurement Distance (m)
30 - 88	100	3
88 - 216	150	3
216 - 960	200	3
Above 960	500	3

Table 4.1 - Radiated emission limits, taken from FCC Title 47, Section 15.209

MHz	MHz	GHz
608 - 614	2200 - 2300	4.5 - 5.15
960 - 1240	2310 - 2390	5.35 - 5.46
1300 - 1427	2483.5 - 2500	7.25 - 7.75
1435 – 1626.5	2690 - 2900	8.025 - 8.5

1645.5 - 1646.5	3332 - 3339	9.0 - 9.5
1660 - 1710	3345.8 - 3358	9.3 - 9.5
1718.8 - 1722.2	3600 - 4400	10.6 - 12.7

Table 4.2 - Restricted frequency bands, taken from FCC Title 47, Section 15.205

The FCC allocates certain unrestricted bands for Industrial, Scientific, and Medical (ISM) use. Two commonly used ISM frequency bands are 902 – 928 MHz, commonly used for short range radio applications, and 2.4 - 2.4835 GHz, which is mainly used for WiFi and Bluetooth. The ISM bands are less restrictive on power levels and duty cycles, allowing significantly higher transmitted power than the general radiated emission limits shown in Table 4.1. Under FCC Section 15.247, compliant systems are allowed to transmit up to +30 dBm (1 Watt) of total conducted output power in any direction. This is a hundred-fold increase in allowable power over the other bands and transmission schemes. While the ISM bands are less restrictive on power levels, it should be noted that harmonics of the fundamental signal that fall in non-ISM bands or even restricted bands must comply with the limits shown in Table 4.1.

When using a single frequency in the ISM band, the maximum permitted radiated power is significantly lower than when utilizing spread spectrum or digital modulation techniques. Under FCC Part 15.427, systems that employ Frequency Hopping Spread Spectrum are allowed significantly higher transmitter output power and Equivalent Isotropic Radiated Power (EIRP), which is a measure of the power transmitted by an antenna in a provided direction. To be allowed this higher power, systems which use FHSS must also transmit across many channels (typically above 50), comply with hopping timings and characteristics, and not exceed regulated out of band transmission levels.

Frequency Band	Transmission Type	Fundamental (EIRP)	Harmonics (EIRP)
902 – 928 MHz	Frequency Hopping	+36 dBm (> 50 Channels) + 30dBm (< 50 Channels)	20 dB below the highest in-band power level
	Single Frequency	-1.23 dBm	-41.23 dBm
2.4 - 2.4835 GHz	Frequency Hopping	+36 dBm (> 75 Channels) + 30dBm (< 75 Channels)	20 dB below the highest in-band power level
	Single Frequency	-0.23 dBm	-41.23 dBm

Table 4.3 - Maximum EIRP for ISM Band Frequencies [x]

As mentioned prior, the gain of the transmitter is limited to a given EIRP level, of which a summary can be seen in Table 4.3. EIRP can be expressed as follows:

$$EIRP\ (dBm) = P_{TX} + G_{ANT} - L_{SYS}$$

In this equation, EIRP is the sum of the transmitter power and the gain of the attached antenna with losses between elements considered. In most PCBs where components are kept close and matching circuits are implemented properly, losses will be near zero. Since high gain antennas focus RF energy into a smaller beamwidth, a greater power density can be sent to a receiver without increasing the power of the transmitter. As such, FCC Part 15.247 stipulates the power of the transmitter must be reduced an equal amount when using directional antennas with gains above +6 dBi. For example, if a +12 dBi directional antenna is used in a FHSS system which transmits across more than 50 channels, then the maximum transmitter power is regulated to +26 dBi.

In addition to power limitations, the FCC also places restrictions upon emissions outside the intended operating band. Harmonic and spurious emissions must not exceed the field strengths listed in FCC Section 15.209, which are summarized in Table 4.1. These restrictions ensure wireless devices do not interfere with other systems which are used for commercial and private applications, such as satellites, TV, radio, and aviation.

4.2 IEEE 802.15.4

The Institute of Electrical and Electronics Engineers (IEEE) Standards Association is responsible for maintaining and overseeing many broad industry consensus standards and their documents. These standards include areas such as power, artificial intelligence, robotics, automotive, and IoT devices. One such standard is IEEE 802.15.4, which is the IEEE standard for the physical (PHY) and medium access control (MAC) layers of low-power, low-data-rate wireless personal area networks (LR-WPANs). IEEE 802 is a family of networking standards that define data transmission protocols for local area networks (LANs), personal area networks (PANs), and other area networks like Ethernet. Within IEEE 802 is 802.15, which is the set of standards that define wireless transmissions for low-power, short-range communications such as Bluetooth and Zigbee. Our project falls under the area of this standard due to its use of low-power wireless communications between devices.

5.0 Application of AI

ChatGPT and other similar artificial intelligence large language models (LLMs), including Claude, Google Gemini, and Microsoft's Copilot, are steadily becoming more powerful tools that fulfill the same role as traditional search engines. Like search engines, LLMs can be prompted using a query to obtain an answer. However, these modern AI tools allow someone to use broader prompts to more quickly obtain either a wider variety of answers or a more nuanced response. The ability to search in a conversational manner is a key aspect of what makes LLMs such powerful search tools, as one can quickly type a question or string of ideas and expect response that covers what was asked and more. Another major advantage LLMs have over the traditional search engine is the length of input query that can be used; search engines typically can only intake a sentence whereas LLMs can have their outputs architected using multi-paragraph prompts.

LLMs have the ability to quickly parse the sources a traditional search engine would show and create a response that synthesizes the information contained within them. This drastically reduces the time it would take someone to familiarize themselves with a topic by skipping the initial search for information across many sources. The difference between search engines and LLMs becomes especially apparent when attempting to digest literature. Typically, researching information from research papers and published books requires using a database which contains these articles and searching with the specific terms you are looking for. Then, you would need to read through each paper or available book to identify the relevant information you are looking for. Using an AI search tool helps to skip this long searching process by presenting you with a summarized result which synthesizes the information across relevant papers and books. The AI answer can also be told to include the sources of its information when it presents its answer to help indicate where information is coming from.

However, the benefits of using LLMs do not come without their downsides, with the biggest of which being possible inaccuracies in its responses. AI search tools are trained on large collections of information, including publicly available and sometimes licensed websites, articles, and books. During this training, the model creates associations between data that can then be later used when generating a response. Since training takes time and must be completed before an LLM can be released as a tool, the databases which they are trained on will never be fully up to date. For example, the most recent ChatGPT version (ChatGPT 5.5) was released in April 2026 but was trained only up to December 2025. In this way, the generative information which these LLMs can respond with can be outdated and inaccurate. However, it should be noted that being outdated only affects the generative, or “thinking”, aspect of the model since it can still access the internet for information on recent events.

6.0 Hardware Design

6.1 Transceiver PCB Design

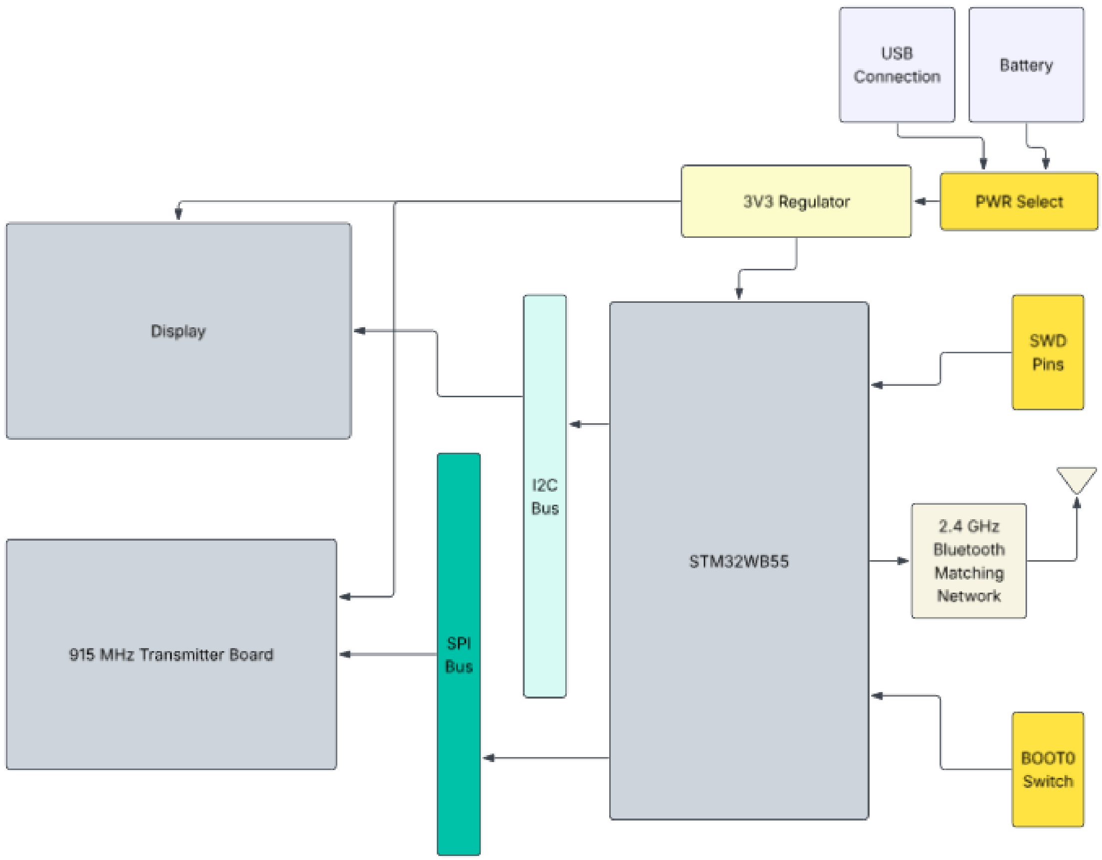

Figure 6.1 - Transceiver Sub-system Block Diagram

The core of the transceiver PCB is the STM32WB55 microcontroller. The chip supports multiple options for voltage power rails, debugging pins, reset, and boot; thus, the user manual and development board were heavily referenced when designing the PCB schematic. Among the power rails there are digital, analog, RF, battery, external regulation, and USB pins which can be used to supply different types of components with varying voltage level needs. Nearly everything on our board can be supplied with 3.3 V, thus the digital, analog, battery, and USB supply pins were connected to the same 3.3 V rail and decoupled with capacitor values per the user manual. The Switched Mode Power Supply (SMPS) and RF voltage rails also operate at 3.3 V, but they require different decoupling capacitor values. As such, these rails are separated from the others and decoupled accordingly.

The STM32WB55 requires two external crystal oscillators: one which operates at a lower frequency of 32.768 kHz and another at a higher frequency of 32 MHz. Of these, only the low speed external (LSE) oscillator requires capacitors to achieve stable oscillation. The

ABS07-32.768KHZ-T was chosen for the LSE oscillator and it has a load capacitance of 12.5 pF. Assuming the FR4 board and traces between the oscillator and capacitors will have roughly 2.5 pF of parasitic capacitance, two 20 pF capacitors are needed to achieve stable oscillation with this oscillator. The BOOT0, NRST, and SMPS circuitry was implemented according to the user manual and component selection was mimicked from STM32WB55 NUCLEO development board.

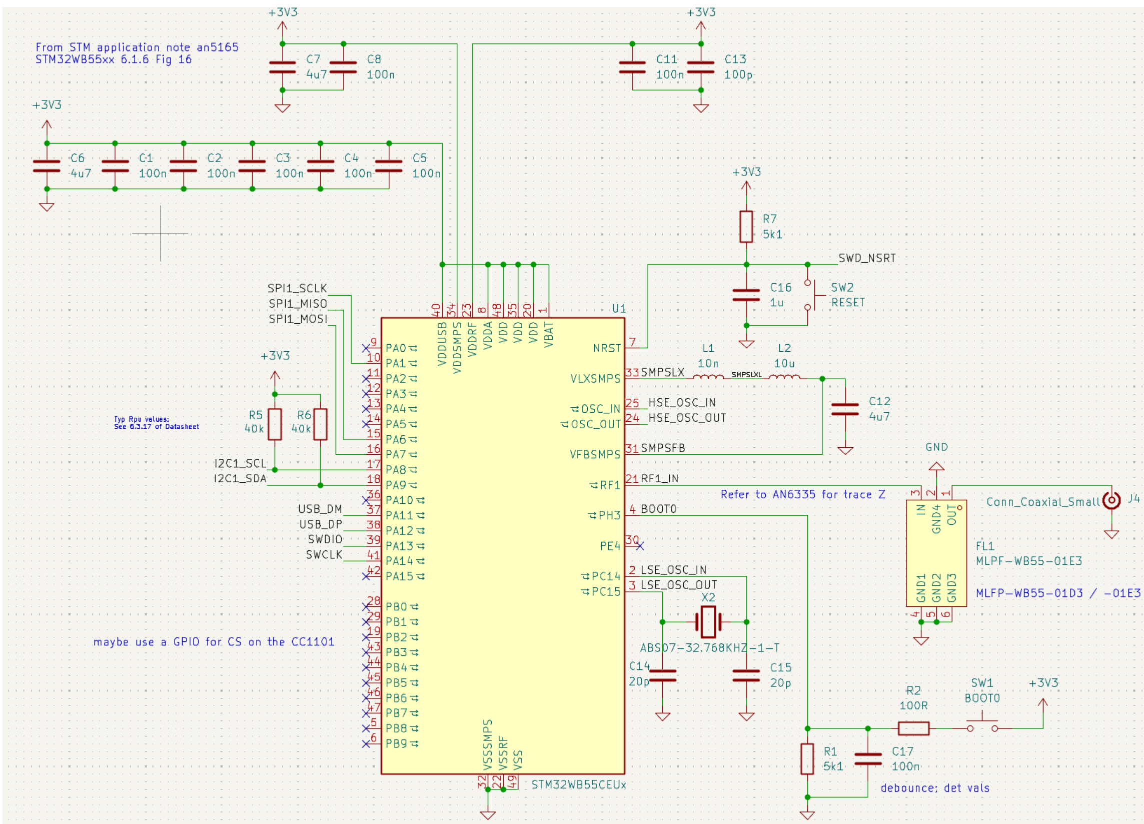

Figure 6.2 - Transceiver Sub-system Schematic

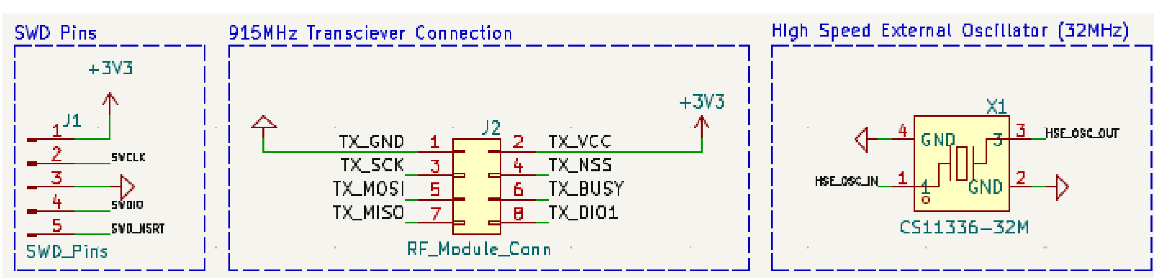

Figure 6.3 - Additional STM32WB55 Connections

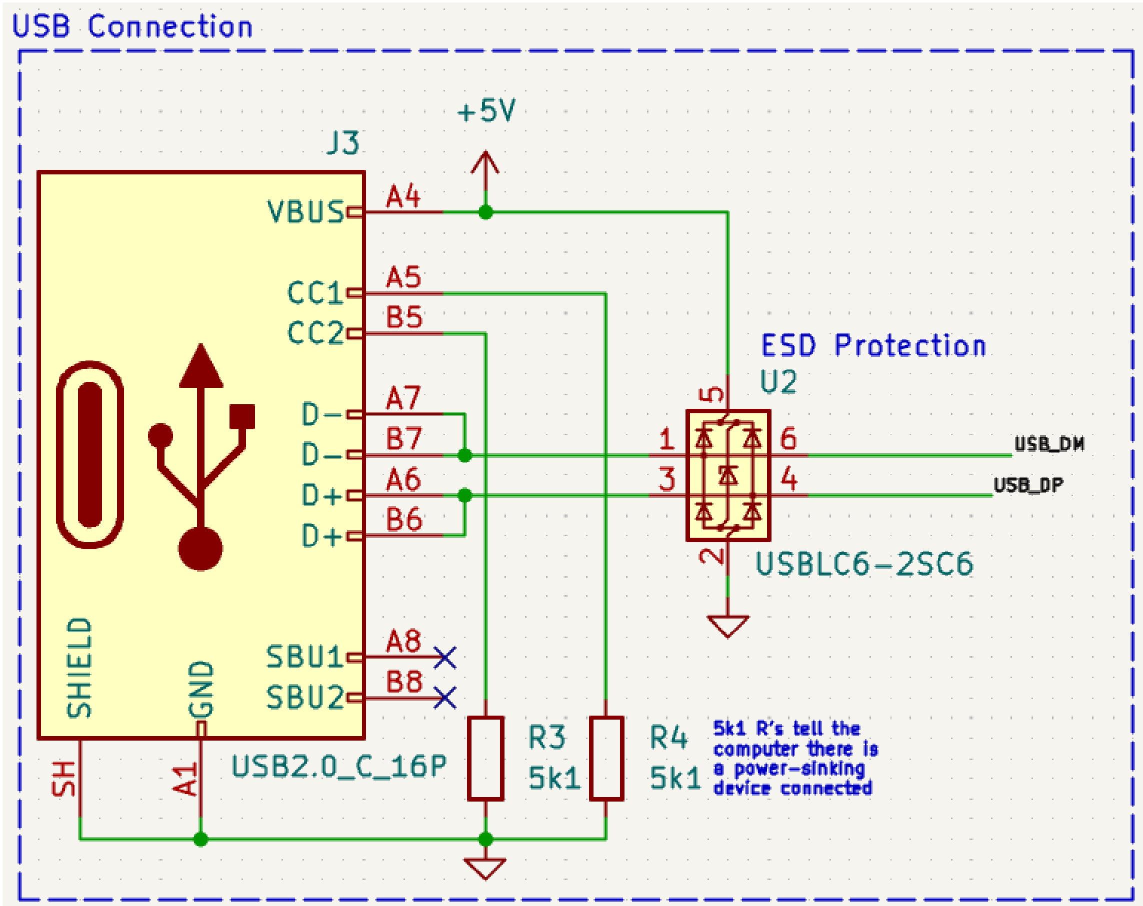

Figure 6.4 - USB C Connection with ESD Protection

6.2 Energy Harvesting PCB Design

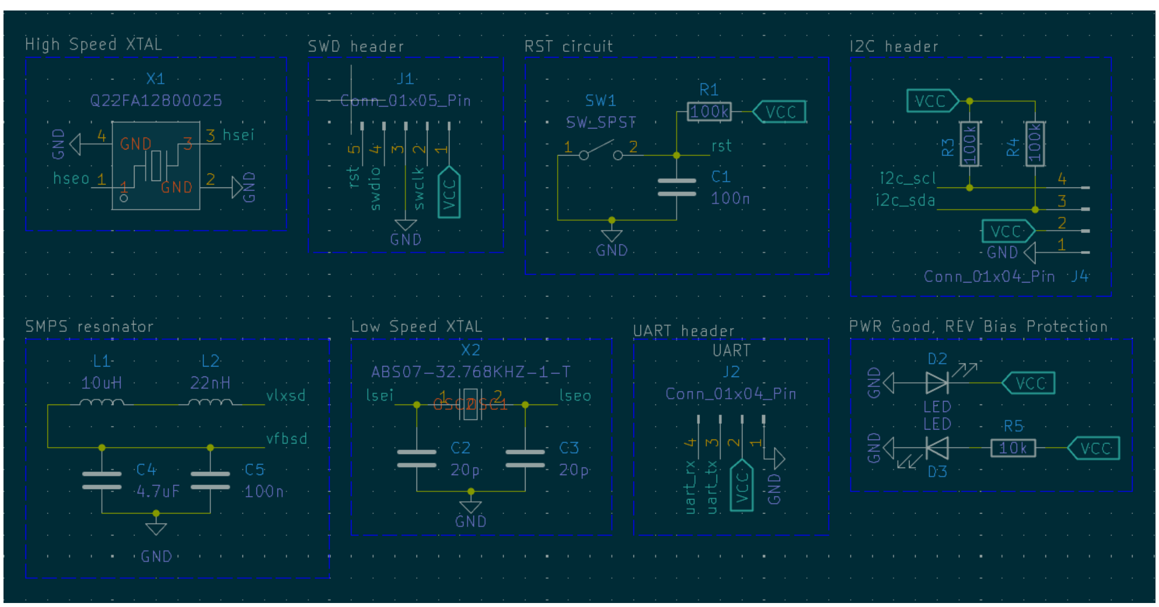

Figure 6.5 - Various hardware setup sections

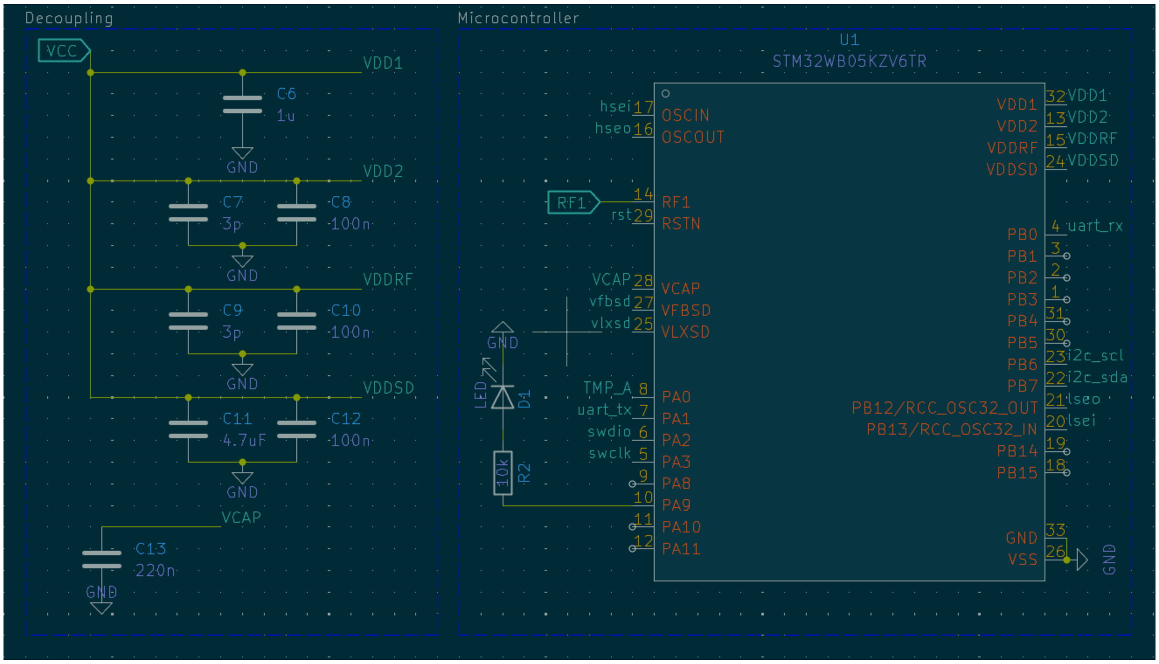

Figure 6.6 - Microcontroller and power decoupling capacitors

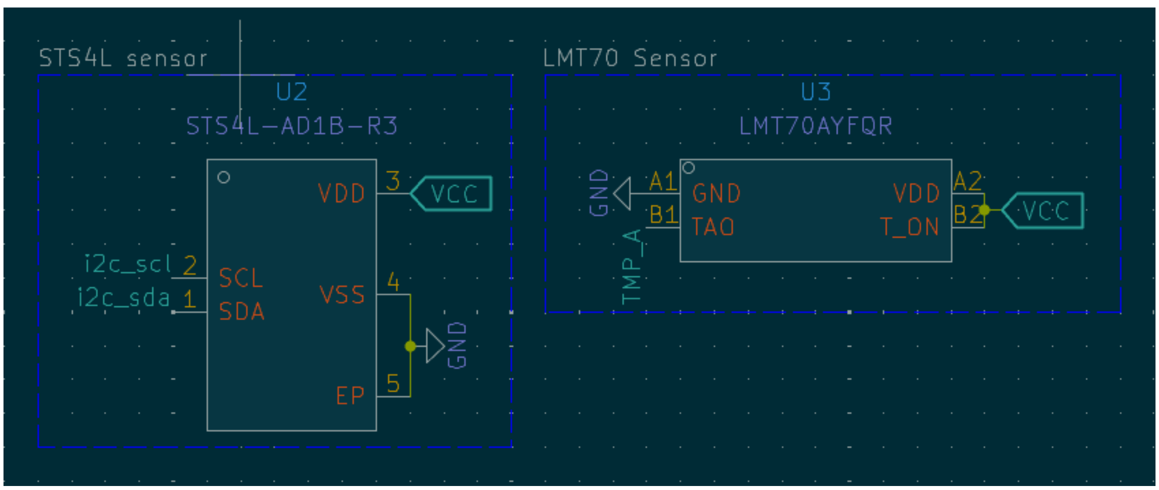

Figure 6.7 - Sensors

7.0 Software Design

7.1 Handheld Transceiver Software Architecture

A key feature of the handheld transceiver is its ability to generate an RF power signal in response to user input. When the user presses a button, the firmware transitions into a controlled RF power transmission routine. This routine is managed by an interrupt that immediately prepares the RF front end for energy delivery. The firmware drives the RF output stage to emit an energy signal toward the environmental sensing board. This will then begin providing the sensor board with energy, allowing it to wake, sense, and transmit updated environmental data. The transceiver carefully regulates the RF power signal to comply with regulatory constraints. While transmitting an RF power signal, the firmware will enable a Bluetooth Low Energy (BLE) scanning state, ready to receive the sensor board's data collection. This button activated mechanism ensures that the user can request up to date environmental data.

Continuing, the handheld transceiver serves as the central coordinating device in the system, responsible for initiating communication, receiving sensor data, and presenting information to the user. Its software architecture is designed around reliability, responsiveness, and interaction with the energy harvesting environmental sensing board. Because the transceiver has a stable power source, its firmware can maintain continuous BLE scanning, perform processing, and support a user interface without compromising system performance.

At the core of the transceiver's software is the BLE central role implementation. The device will continuously scan for data packets broadcast by the environmental sensing board. Once data is detected, the transceiver initiates a connection request. This connection process is managed by a dedicated BLE manager module that handles pairing, connection intervals, and data throughput optimization. The BLE manager also ensures that the transceiver handles disconnections, timeouts, and corrupted packets.

After establishing a connection, the transceiver transitions into a data acquisition state. In this state, the firmware reads characteristic values from the sensor board. These characteristics will include temperature and humidity. The transceiver will validate each received packet as validation ensures that environmental data is not corrupted by interference or incomplete transmissions.

Once data is validated, it will then begin processing. While processing, the data will then be converted into meaningful descriptive engineering units and/or applying calibration techniques to ensure accuracy. For example, temperature values may be corrected using a linear calibration model derived during system characterization, which will be especially important for analog temperature sensors. The electronic display will then present processed data to the user through the transceiver's display. This will also include status for BLE connectivity and battery level. The device will primarily respond to button presses and sensor data retrievals.

Error handling is another critical component of the transceiver's software design. The firmware monitors BLE connection quality and packet integrity. If communication fails, the transceiver automatically switches back to scanning mode. Diagnostic messages are displayed to the user when continued issues occur.

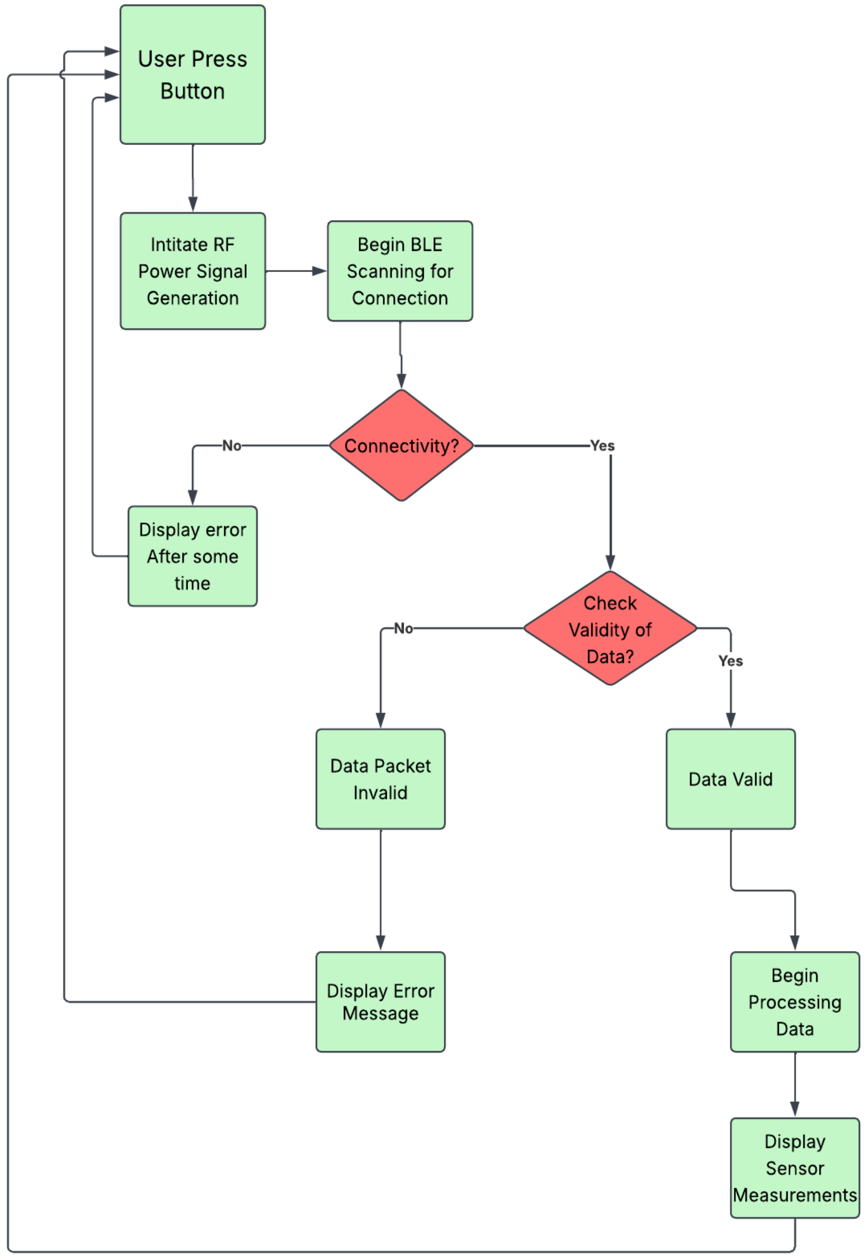

Figure 7.1 - Handheld Transceiver Software Flowchart

7.1.1 BLE - Bluetooth Low Energy Communication Design

7.1.2 RF Power Signal Transmission Control

7.1.3 User Interface and Input Handling

7.1.4 Data Processing and Calibration

7.2 Environmental Sensing Board Software Architecture

The software architecture for the environmental sensing board is optimized for ultra-low-power operation. Because the system relies on RF energy harvesting, the firmware must operate under tightly constrained and limited power availability. This demands a design that must be focused on short, efficient execution bursts, and sleep scheduling. The architecture is divided into sensing, power-management, data-processing, and wireless communication levels, each designed to operate independently.

At the sensing layer, the firmware interfaces with two distinct environmental sensors: the analog LMT70 temperature sensor and the digital SHT41 temperature/humidity (for our purposes, Humidity sensing only) sensor. The LMT70 requires an ADC sampling routine with calibration logic to convert its analog voltage output into temperature readings with sharp precision. In contrast, the SHT41 communicates over I2C, so the software will include a compact driver that handles initialization, measurement, and CRC-validated data retrieval.

Power management is perhaps the most critical part of the architecture due to the RF energy harvesting subsystem. The firmware continuously monitors available energy using a voltage thresholding mechanism. When harvested energy is low, the system enters a deep-sleep mode, waking only periodically to check whether sufficient charge has accumulated. When energy is adequate, the scheduler activates sensing and communication tasks. This adaptive duty cycling ensures that the board never attempts high power consumption operations.

Data processing will focus on turning sensor readings into small, efficient packets for BLE transmission. Because the board runs on limited harvested energy, the firmware uses a compact format to keep radio use as short as possible. Measurements are stored in a small amount in memory, so the device can hold onto recent data if it doesn’t have enough power to transmit right away. Once energy levels rise, the stored readings are sent out in the next available BLE broadcast cycle.

Finally, the BLE communication handles broadcasting, connection management, and packet transmission. Because BLE will be one of the most energy-intensive subsystems, the architecture will attempt to broadcast sensor data without requiring a persistent link. When a connection is needed, for example, during configuration or diagnostics the firmware enables a minimal GATT service with data for temperature and humidity transmission. The BLE stack is configured for low-duty-cycle operation and complete shutdown during energy scarcity. Together, these strategies allow the sensing board to reliably transmit environmental data despite relying solely on harvested RF energy.

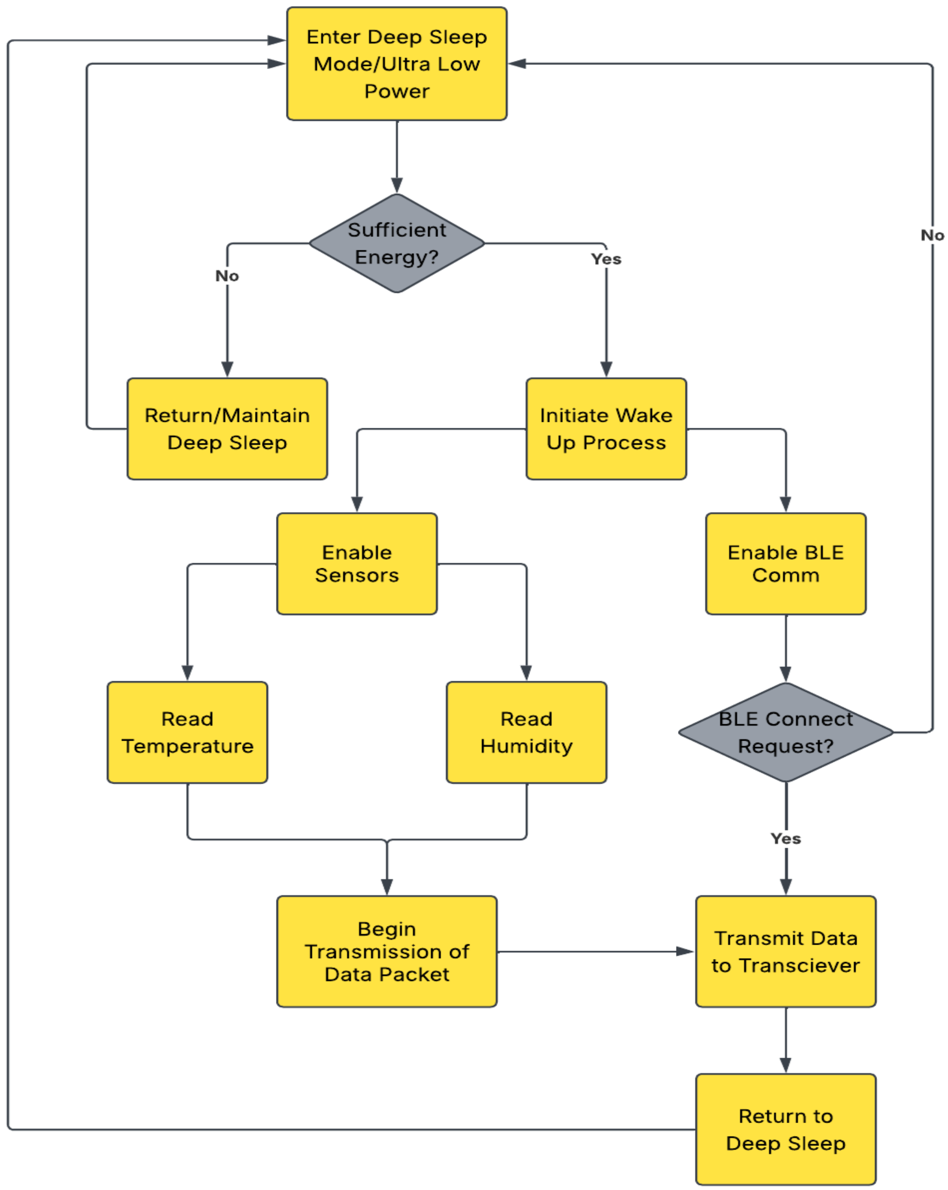

Figure 7.2 - Environmental Sensing Board Software Flowchart

7.2.1 BLE - Bluetooth Low Energy Communication Design

7.2.2 Environmental Sensing Firmware Implementation

7.2.3 Power Management and Duty Cycling

7.2.4 RF Energy harvesting Monitoring

7.2.5 Data Packaging and Transmission Format

10.0 Administrative Content

For this project, our group came up with an initial Budget and Bill of Materials. An itemized list of the materials required for the project, their costs, and the quantity needed is shown in Table 3. The exact price and material usage of the project will be noted once it is completely designed and fabricated.

Table 10.1 - Budget

Category	Budget	Information
Integration and Testing	$200.00	Includes items such as development boards, prototyping PCBs, and breadboard components
PCBs and Electronic Components	$200.00	Includes procurement of PCBs, shipping costs, and discrete electrical components
Total	$400.00	

Table 10.2 - Bill of Materials

Bill of Materials				
Item	**Name**	**Price**	**Quantity**	**Details**
MCU	STM32WB05KZ	$5.00	2	MCU Sensor Board
MCU	STM32WB55	$16.00	2	MCU Handheld Device
Development Boards	NUCLEO-WB05KZ	$25.59	1	Debugging and Development
	P-NUCLEO-WB55	$80.00	1	
Temp Sensor	LMT70	$2.64	1	Temperature Measuring
Humidity sensor	SHT41	$3.50	1	Humidity Measuring
	Semtech SX1262	$10.00	1	RFIC

915 MHz Transmitter	Waveshare SX1262 Development Board	$16.00	1	Development Board for Integration and Testing Purposes
OLED/LCD	Elego 0.96 Inch OLED Display	$10.00	3	User Display

To keep pace with our project, we developed a table of project milestones with tasks that need to be accomplished by a certain time. This table details anticipated time frames for each part of the project and designates the group member(s) who will be responsible for a given task. It shows the project milestones that are part of a two-semester long project (SD1 and SD2). Our group created a list of milestones to ensure our team will be using our time wisely. We sought to maintain steady progress on all project objectives to prevent complications arising from the project's strict time limitations. Project milestones can be found in Table 10.3.

Table 10.3 - Project Milestones

Senior Design I - Summer 2026			
Task	**Time Period**	**Dates**	**Responsibility**
Project Selection	1 Week	5/14 - 5/21	All
D&C Document	1 Week	5/21 - 5/29	All
D&C Meeting	30 Minutes	6/1	All
Research Transceiver	2 Weeks	6/1 - 6/15	Jacob
Research Antenna Design	2 Weeks	6/1 - 6/15	Jacob
Research Energy Harvesting	2 Weeks	6/1 - 6/15	Ryan
Research RF Rx to DC conversion	3 Weeks	6/1 - 6/22	Ryan
Research Sensor Integration and Power Usage	2 Weeks	6/8 - 6/22	Ryan
Research MCU Power Management	3 Weeks	6/1 - 6/22	All

Research Programming of MCU	3 Weeks	6/8 - 6/29	Nicholas
Research Data Tx to Rx Transceiver	3 Weeks	6/8 - 6/29	Nicholas
Research Display of Sensor Data	2 Weeks	6/15 - 6/29	Nicholas, Jacob
Start PCB Design	2 Weeks	6/29 - 7/13	All
Final Decision on PCB Design	1 Week	7/13 - 7/20	All
Start Project Prototype	2 Weeks	7/13 - 7/28	All
60 Page Draft	2 Weeks	6/22 - 7/6	All
Final Documentation	5 Weeks	6/22 - 7/28	All
Senior Design II - Fall 2026			
Task	**Time Period**	**Dates**	**Responsibility**
PCB Testing	4 Weeks	TBD	All
System Level Testing	4 Weeks	TBD	All
Project Demo	1 Week	TBD	All

Appendix A – References

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Appendix B – Copyright Permission

Appendix C – Data Sheets (if necessary)

Appendix D – Software Code (if necessary)

Appendix E – ChatGPT prompts and outcomes (if necessary)